R&D Support Project Proposal for SMEs Technology Development

❶ Project Overview

Development of RAG-Based Business Specialization Real-Time Multilingual AI Translation Solution - Global Sales, Consultation, Translation, Document Automation Integrated Platform for SME Export Sales Innovation

Project Name

Implementation of AI models specialized for business: Develop a RAG-based multilingual translation model trained on unique company knowledge (product names, technical specifications, prices, etc.) and an End-to-End (E2E) real-time streaming voice interpretation model (in more than 8 languages) with a delay of within 2 seconds.
Automation of the entire export business cycle: Develop 5 types of export document auto-generation agents such as quotes and product pages.
Unique regulatory certification formatting: Build a SaaS platform integrating complex regulatory certification document formatting modules such as eCTD, KC, CE, FCC to remove global entry barriers for SMEs and achieve sales innovation.

Development Goal

[Core Differentiation Data] Establish an overwhelming entry barrier by conducting domain-specific fine-tuning using an exclusive corpus with 1.1 million sentence pairs (600,000 business + 500,000 regulatory certifications) that cannot be replicated in a short period.
[AI Model Advancement] Combine RAG technology that injects company-specific knowledge into a hybrid language model based on NLLB/EXAONE to reduce hallucination phenomena by 70%. Secure simultaneous interpretation performance within a 2-second delay with InfiniSST and Whisper-based technology.
[SaaS Platform Commercialization] Develop a cloud SaaS web, mobile, API integrated environment that allows demand companies to link and use their data within 3 days without IT personnel, utilizing the verified AI platform (Oz-i) infrastructure of the main organization.

Development Method

[Main] PCN Co., Ltd. (Platform/Commercialization General Manager): 25 years of operation (annual sales of 27 billion won), an AI company with a GS-certified platform, possessing the Oz-i AI platform and extensive know-how in various R&D projects, ensures the construction of the SaaS platform, API integration, and cloud infrastructure commercialization.

[Collaborator①] StunningBox Co., Ltd. (AI Model/Data Specialist): An AI translation data company with 15 years of accumulated professional datasets, responsible for AI translation fine-tuning, providing its core asset of an exclusive corpus, and multi-agent regulatory certification formatting technology.
[Collaborator②] 3 Demanding Companies (Field Proof Verification):
Sunlight Illucom Co., Ltd. (automotive parts), Genuine Grip Co., Ltd. (special shoes), Grey Whale (K-Beauty), all actual top export award companies, provide proof data for three industries and quantitatively verify the introduction effects through A/B testing.

Research Team Composition

[Economic and Industrial Impact] Enhance the export contract rate of demand companies by over 20% and drastically reduce the regulatory certification document processing period from the current 4–8 weeks to 1–2 weeks. By the fifth year of commercialization, create 22 billion won in sales (cumulative 68 billion won over 5 years), generate 51 new jobs, and plan to supply to over 500 SMEs by linking AI voucher and KOTRA. Achieve a cumulative 44 billion won export substitution effect over 5 years.
[World-Class Technology Acquisition] Achieve translation accuracy with BLEU over 65, voice interpretation delay within 2 seconds, and document structure retention (TEDS) over 70%, to preempt the B2B domain-specialized translation and interpretation market that global big tech (Google, DeepL) could not solve.

Expected Effects

2 Self-Assessment

Project Name

Not Applicable

Applicable

Checklist

1. Does the current applicant (R&D organization, etc., hereinafter referred to as 'applicant') not meet the application qualifications specified in the business announcement, etc.?

2. Are there any documents among the related documents stipulated by the business announcement, etc., that the current applicant has not submitted?

3. Is the current applicant subject to national R&D participation restrictions?

4. Does the applicant have a history of sanctions, such as restrictions on government R&D participation due to misuse of labor costs or fraudulent transactions?

5. Is the current applicant a bankrupt, suspended, or closed company?

6. Is the current applicant under tax delinquency disposition by the tax authorities for national tax, local tax, customs duties, etc., or registered with a comprehensive credit information concentration agency for fine delinquency, industrial accident/employment insurance delinquency, etc.?

7. Is the current applicant registered in the debtor list based on the Civil Execution Act or as a debtor with a comprehensive credit information concentration agency?

8. Has the current applicant filed for the commencement of bankruptcy, rehabilitation procedures, or individual rehabilitation procedures in court?

9. Based on the most recent year's financial statements, does the recent applicant have a debt ratio of over 1000% or is fully impaired in capital?

10. As a result of its own inspection, does the application task fall under the security task classification criteria of Article 45 Paragraph 1 of the Enforcement Decree of the Innovation Act as a security task?

※ In addition to the items specified in the self-assessment table above, support may be excluded or selection may be canceled depending on the review results of the application qualification criteria prescribed by the business announcement and related regulations
※ The criterion of 'current' in the checklist above is the point set for confirming the application qualification requirements in the business announcement, related regulations, etc., such as the application deadline
※ Each main and joint institution needs to submit

This applicant has confirmed the facts regarding the above items, and does not object to the possibility of support exclusion or selection cancellation if they differ from the facts.

Name

Year 2026

July

IAL RS

Main Institution Applicant: Sang-hyun Yeom

Main Institution Corporation Name: CN Co., Ltd.

1. Technology Development Goal

Standardize the development of RAG-based business-specialized multilingual AI translation solutions as cloud SaaS to develop an integrated platform that allows any SME to link its own data for the automation of global sales, interpretation, and documents.

Final Goal

StunningBiz AI Translation Platform (based on PCN Oz-i)

SME Applicable Industries

SaaS Utilization · Spread Across All Industries

Manufacturing

SaaS Subscription Service

StunningBiz AI Cloud SaaS Marketplace

10,000~50,000 won/month Subscription Knowledge Data RAG Linking

Automotive Parts, Electronics, Machinery, Textiles

RAG Knowledge DB

AI Translation Engine

Industry-specific Training Data

Corpus and Terminology Dictionary

- Wholesale and Retail Trade

Company-specific Vector DB

Exemplary Serving - Front

API Integration (B2B2B)

Export Trade, Distribution, E-commerce

Cafe24 · Amazon · Naver

CRM · ERP · Shopping Mall Integration

AI Translation Data Pipeline

· Food · Bio

Regulatory Certification Premium

Solution 1

Health Functional Foods, Cosmetics, Pharmaceuticals

Solution 2

Solution 3

eCTD·CE·FCC·HACCP exclusive billing (50% ↓ compared to existing)

E2E Real-time Cooperative Customer Engagement

Document Automation Type\n- Regulatory Certification

Business Model Connection

- IT · Service Industry

□ Spread Across All Industries

Periodic Re-learning Based on CI/CD · Company-specific FAQ Auto-update · Model Performance Improvement Automation

SW, SaaS, Content, Experience

KOTRA·Trade Association·AI Voucher SME 500+ Distribution

Real-time Interpretation Service

Document Auto-generation

Custom RAG Translation

· Agriculture and Fisheries · Others

SME Application Scenario

K-Food, Agricultural Export

Quotations · Catalogs
Contracts · Regulatory Certification

8+ languages upper direction
Operation within 2 seconds

Corporate terminology unified 40%↑

Correct reflection of product name · price
Industry-specific terminology

Real-time buyer consultation interpretation
Automatic multilingual shopping mall construction
Automatic consistent generation of export documents

Speaker separation · Subtitles
Zoom/Teams integration

Automatic response to overseas inquiries

Integration with company's own data

70% time reduction

Achievement of BLEU 65+

Product name · Glossary · Price list

Web · Mobile App · Zoom/Teams plugin · Shopping Mall API (Cafe24·Amazon·Shopify) · CRM/ERP integration · Offline exhibition mode

PC1 Oz-i

Starbox

Key Performance Indicator (KPI)

AI platform integration

Standard·Certification code issues

Proposal 2 trillion ↓

TEDS 70%+

BLEU 65+

F1.08+

WER 15% ↓

500,000 military drivers

Perfect SaaS infrastructure (Revenue 28 billion · 200 employees scale)

Translation accuracy

Real-time interpretation

Term extraction

Speech recognition

Document structure preservation

(Barrier to entry · Initial capital gap)

Commercialization 1st year sales 6 billion won | 5th-year sales 22 billion won, employment of 51 persons | Export conventions 5-year cumulative approximately 44 billion won

1) Background and need for business promotion

(1) Environmental analysis

Export SMEs face complex barriers in overseas sales processes due to translations that do not reflect company-specific contexts, interpretation structures unsuitable for real-time consultation, and the high cost and long duration of regulatory certification document handling. Thus, beyond simple translation functions, a domain-specific AI platform that connects the entire export business cycle based on company data and specialized terminology is needed.

Key barriers

On-site issues

Need for technology development

① Work context·Specialized terminology

Generic AI fails to sufficiently reflect product names, technical specifications, prices, and contract terms per company, leading to a risk of critical information mistranslation during negotiation and contract processes.

Integrate each company's product DB, price list, and glossary with RAG to reflect in real-time in translation, interpretation, and document creation.

② Real-time interpretation performance

The existing 3-stage structure of STT→MT→TTS accumulates a delay of over 2-5 seconds, with limits on background noise, speaker separation, and specialized terminology processing.

Realize interpretation within 2 seconds through E2E streaming interpretation architecture, speaker separation, and specialized terminology injection.

③ Regulatory certification documents

CE·eCTD·FCC·HACCP, etc. have complex national formats and document structures, posing a significant burden of preparation time and outsourcing costs on SMEs.

Automate translation, structuring, and formatting of regulatory documents and support submission document preparation based on expert review (HITL).

(2) Market growth and development urgency

◦ The global business AI translation market is projected to grow from about $1.8 billion in 2025 to over $7 billion in 2030, with rapidly expanding demand for business domain-specific solutions.

◦ The regulatory certification service market sees substantial costs in document preparation, translation, and formatting, increasing the demand for time and cost savings through document automation.
◦ There is a market opportunity for a SaaS-integrated solution that bridges the gap between general translations and high-cost specialized services for domestic export SMEs.

To connect these market and field demands to actual export performance, improving only the accuracy of translation is not enough. It is necessary to integrate unique company knowledge in real-time translation and interpretation and to link consultation, document creation, and regulatory certification responses into a single workflow. Therefore, this project aims to develop a domain-specific AI platform supporting the full cycle of export business based on company data and specialized terminology.

2) Limitations of existing solutions and the direction of this business establishment
(1) Limitations of existing solutions
The limitations of existing solutions lie not only in the insufficient performance of individual functions but also in the structural problem where business context, real-time consultation, document, and certification tasks are disconnected. This project will automate the entire export business cycle by connecting it based on corporate RAG and domain-specific models.

Unresolved with just translation function! — A platform that connects the full cycle of export business is necessary

AS-IS

Type 2 real-time performance not guaranteed

Type 1 non-execution of work commencement

Type 3 undeveloped conditional items

eCTD-CE HACCP international format standards are the conclusion

Accumulated tasks of STT, MT, TTS

Translation errors + Negotiation missteps lead to contract risks

Failure wisdom at 2~5 trillion is impossible to rival.

Not tuning but architecture issue

+ First Mover opportunity

TO-BE

Business Model

Document

General

Person

Keep RAC

It's real-time

<CTBOP HAKOP>

Translation document

Childcare

Child's admission

Production product's
Appraisal contract

Expense period
Main administrative positions

Incoterms Duty

Delayed Toss Wa

Friend Come Out

② Affiliate Oil Location

Resolution of Natural Structures

RAC Basic Documentation

(2) Final Development Objective and Three Major Solutions of this Project

Three Major Solutions

Key Development Content

Main Objectives

① Business Translation Model

NLLB/EXAONE-based Domain Learning + Per-enterprise RAG Knowledge DB + Expert Feedback Learning

BLEU 65+
Terminology Consistency 40%↑
Hallucination 70%↓

② E2E Real-time Voice Interpretation

Whisper·InfiniSST-based Streaming Interpretation, Speaker Separation, Real-time Enterprise-specific Terminology Injection

8+ Languages
Delay Less than 2 Seconds
WER 15%↓

③ Automatic Document Creation· Regulatory Certification

Development of 5 Types of Export Document Agents and eCTD·CE·FCC·HACCP Formatting Modules

TEDS 70%+
Document Creation Time 70%↓
F1 0.8+

(3) Development Direction

This project does not stop at developing the three solutions individually but integrates the domain-specific translation model and enterprise-specific RAG knowledge DB as a common foundation. Through

this, the product information and specialist terminology of enterprises are consistently reflected in real-time interpretation and document creation, automating the full operation cycle from overseas sales, buyer consultations, export document creation, to regulatory certification response, on a single AI agent platform.

3) Status of Domestic and Overseas Technology and Competitive Environment

While solutions exist in translation, interpretation, and document processing in domestic and international markets, there is limited case of providing sales,
consultation, real-time interpretation, professional translation, and regulatory document automation as a unified SaaS for SMEs.

(1) Major Technology Status at Home and Abroad

Leading Companies·Services	Function Area	Main Strengths	Limitations from Project Perspective
DeepL	Translation	Excellent general text translation quality and user-friendliness	Real-time reflection of enterprise-specific terms, prices, and contract conditions, creation of regulatory document structure, linking of business processes is limited
Google Gemini Live Translate	Real-time Interpretation	Low-latency multilingual interpretation feature focused on everyday conversations	Consistency of business-specific terminology, management of multiple speakers, linkages with enterprise documents·RAG is limited

High-cost, long lead-time
structure based on human resources, with limited
automation as a SaaS for SMEs and new document creation features

Professional Translation
Service

TransPerfect·RWS

Professional personnel and multilingual
project operation capability

Lacks features in export regulation domain, creation of new multilingual documents,
and enterprise-specific terminology RAG

Document Processing·
Automation

Hyperscience·UiPat
h

Capabilities in unstructured document classification·extraction and business automation

Limited generation of regulatory documents, regulatory DB,
and linkage of the overall export business cycle

Domestic Translation

Flitto

Leading domestic translation platform
combining AI and human translation

Domestic
Support·AI

KOTRA
Export Voucher,
CLOVA·Saltlux·Ma
indslab, etc.

Possession of translation support or AI
documentation·consultation capability

Dependency on human resources or limited proliferation of SaaS specialized for export·trade·regulatory certification and targeting external SMEs

(2) Comparison by Function Area Coverage

Platform

Sales·Consultation

Real-time interpretation

Professional Translation

Regulatory Document Automation

Overall

StunningBiz AI
(This Project)

O

O

O

O

Integrated Offering

Translation-focused

DeepL

X

X

O

X

General-purpose Focused

Google

△

O

△

X

Human Services

TransPerfect·RWS

X

X

△

△

Document Processing-focused

Hyperscience·UiPath

X

X

X

△

Partial Support

Flitto·Domestic AI Solutions

△

△

△

X

※ O: Dedicated feature provided / △: Partial support or general-purpose feature response / X: Not supported. This qualitative comparison is based on publicly available materials, and the final submission will indicate the survey reference date and sources.

(3) Conclusion of Competitive Environment Analysis

The comparison shows that general-purpose translation quality, domain expertise in professional translation services, and the automation strength of document processing solutions
all exist independently. However, these services have limitations in providing a unified SaaS environment that interlinks enterprise-specific product information and terminology through RAG, extending from sales,

consultations, and real-time interpretation to professional translation and regulatory document automation.

Thus, there is a market gap in the domain of integrated AI SaaS for SMEs that supports the full-cycle export operations based on enterprise-specific knowledge.

4) Differentiation·Originality and Technology Development Direction

The uniqueness of this project lies in connecting the workflow from sales→consultation→documentation→certification based on a common foundation of over 1.1 million expert corpora and enterprise-specific RAG knowledge
DB, rather than a simple integration of individual functions.

Owned Assets・Core Technology

Differentiation Axis

Project Implementation Direction

① Exclusive Expert
Data

Over 15 years of accumulated expert parallel corpora
1.1 million+ sentence pairs
- Business: 600,000
- Regulatory certification: 500,000

Leveraging domain learning in trade, sales, contracts, and regulatory certification, and building industry-specific terminology DBs to establish data entry barriers that are difficult to replicate in a short period.

Inject foundation data in real-time for translation, interpretation, and document creation to reduce key errors and hallucinatory issues such as product names, figures, and Incoterms.
Implement to reduce the accumulated delay of the existing three-stage Cascade and to achieve interpretation within 2 seconds during multilateral buyer consultations and exhibitions.
Develop a service that can link corporate data within at least three days and expand to shopping malls, CRM, ERP, and meeting platforms.

② Enterprise Context-Specific
RAG

Product DB, Price Lists, Contract Terms, Translation Memory, and Hybrid Search (BM25+Dense Retrieval)

③ E2E Real-time
Speech Interpretation

Streaming structure based on Whisper and InfiniSST, speaker separation, VAD and noise response

④ Export Business Integrated
SaaS

5 types of export document agents, eCTD, CE, FCC, HACCP formatting, Web, Mobile, API

(1) Direction for Verification and Commercialization Connection

Through field testing by three demanding companies, we verify not only translation accuracy but also the potential for reducing work time, improving document quality, and enhancing export response speed and order acceptance rates.
◦ AI outputs are provided as drafts, and regulatory certification documents are subject to mandatory Human-in-the-loop (HITL) review procedures to manage legal and business risks.

◦ Combining PCN's SaaS, API, and cloud operational capabilities with StunningBox's AI model, data, and regulatory document capabilities to simultaneously demonstrate and commercialize within 24 months.

This project aims to resolve the gap between generic translation AI and high-cost specialized services, based on expert data and enterprise-specific RAG, and develop and demonstrate a B2B domain specialized AI SaaS that integrates the entire export business cycle.

2. Research and Development Method

1) Data Acquisition Method

This dataset is constructed based on 15 years of accumulated real transaction parallel data obtained through an official data utilization contract between the joint R&D entity, StunningBox Inc., and the 'Korean Translation Institute'. It consists of outputs from actual professional translation projects carried out for various SMEs. The raw data provider's security regulations and the Personal Information Protection Act are strictly adhered to, with de-identification and anonymization completed. After that, StunningBox's

proprietary data refinement technology was used to perfectly reprocess and assetize the dataset for AI training. The data utilization contract for this project's execution and the de-identification measures have been reviewed and completed.

Data Acquisition Method — StunningBiz AI Translation and Interpretation Platform
1. StunningBox's own business and regulatory parallel corpus [Core Differentiation - Entry Barriers]
Business collaboration and negotiation dialogues, product descriptions, catalogs, proposals, clinical, regulatory, and certification documents, contracts, trade documents
Korean-English parallel 300,000 sentence pairs • cCTD-KC-CE-FCC 500,000 sentence pairs approximately 100,000 sentence pairs
Total: Over 1.1 million sentence pairs possessed (15 years of accumulated expert dataset) — Exclusive data that competitors cannot easily replicate in a short time
2. Field data collection for three demanding companies - Industry-specific empirical data acquisition
Sunlight Lucorn [Automotive Parts] HB [Footwear Manufacturing] Grey Hear [K-Beauty]
English · German · Japanese English · Chinese · Vietnamese English · Chinese · Japanese
Technical documents · Negotiation dialogues CE · FDA 43 documents
BOM, OEM contracts, etc. catalogs K-Beauty marketing content
Exhibition scripts raw material regulatory agency documents
3 industries × Actual collected astronomical data — Foundation data for building industry-specific RAG simulators — Basis for expanding into all industries
Data refining · De-identification RAG Vector DB embedding AI model training · Fine-tuning · Internalization
Joint construction of industry-specific terminology DBs (Terminology DB) tailored to demanding companies, design of translation · interpretation quality feedback loop, and Active Learning application
Additional collection of public data: KOTRA overseas market news, Korea Customs Service HS code DB, International Certification DB (FDA CDER, EMA EudraLex, KS/ISO)
Crawling and refining multilingual parallel data of product descriptions from overseas e-commerce sites (Amazon·Alibaba·Rakuten), and collecting integrated data linking domestic e-commerce sites
(1) Data Quality Management Plan
a. Core Data Validation Pipeline
Build learning and verification datasets based on the scale of learning data, expert review and effect verification
Data Refining
Over 1.1 million 15-year accumulated expert dataset
Negotiation and Collaboration Dialogues 300,000 catalogs and proposals 200,000
Collected Data Regulatory and Certification Documents 500,000 Contracts and Legal 100,000
KOTRA-Korea Customs Service HS Code International Certification DB
Selected Parser, De-identification Blocking, Labeling, Vector DB, Governance (IAA 0.85+)
RAG Vector • Feature/Label Dataset DB construction / Proposal Evaluation
AI Model Training · Fine-tuning RLHF
Domain Fine-tuning Quality Evaluation and Verification Feedback Loop Labeling Guidelines Learning Verification
Expert Review and Effect Verification
Expert Evaluation (MQM - Field)
Translation Quality Evaluation : BLEU Measurement
Machine Translation Performance (WER)
Negotiation and Legal Verification
Regulatory Document Compliance
Effect Measurement (A/B Testing 3 times)
Artificial Intelligence Market
Daily 2 AI2+ 3,500 Linkage Consistency Test
KOLAS BLEU-WER Accredited Verification
Feedback Loop Expert Review & RLHF Training: Monthly Client Provision

Execution Contents

Stage

① Collection

Collection of Source Data by Demand Company

② Refinement
MinHash-based duplicate removal, Rule-based cleansing (ungrammatical errors, typos filtering)

③ Verification
Sample cross-review by professional trade and translation personnel, anomaly detection

95%+ Accuracy, Duplication Rate below 5%, and Error Rate below 3%. Only pass data (Gold Dataset)
④ Learning Reflection
Incorporate into Fine-tuning

(2) Legal Responsibility and Expert Review of Automated Regulatory Documents (HITL)
HITL Review Procedure
AI Draft Generation → Expert Review → Approval/Rejection → Final Issue (issuance of the final version is not possible without review)
Accountability, Notification, Update System

Item

Content

Legal Responsibility Structure

Clearly notify that AI outputs are at the draft status, and the final submission responsibility lies with the user (demand company).
Insert the domain expert translation reviewer step as a mandatory workflow.
Check updates to country-specific export regulation and customs guidelines at least once a quarter and update rule sets.
Post notification phrases within the service and prepare exemption clauses in the contract.

Expert Review Procedure

Guideline Update by Country

User Notification and Disclaimer Structure

(3) Data Security • Four-layer Structure

Stage

Execution Contents

① Encryption at Rest

Encrypt DB/File Stored Data with AES, etc.

② Encryption In Transit

TLS-based encryption of transmission sections

③ Access Authority Control

Role-Based Access Control (RBAC)

④ Audit Log Record

Track the history of access and changes meticulously

Model Development Method

(1) Business Hybrid Translation Language Model (3-Stage Training Pipeline)

Stage

Method

Core Content

Stage 1
Domain
Fine-tuning

Transfer Learning with Business Corpus

Internalization of domain language patterns based on trade, sales, and contract business parallel corpora (500,000 sentence pairs).
Industry-specific specialized learning for three demand company industries.

Stage 2 RAG
Integration

Enterprise-specific Knowledge
DB Real-time
Search Augmentation

Store unique product names, prices, and conditions in vector DB. During translation/interpretation, inject real-time context via Hybrid Search (BM25+Dense Retrieval). Aim to reduce hallucination by 70%.

Stage 3 RLHF

Expert Feedback Preference Learning

Reinforcement learning based on feedback from translation experts and sales practitioners. Optimization of business negotiation style, honorific speech, and cultural nuances.

※ Of the 600,000 business parallel corpus sentence pairs possessed, a training subset of 500,000 sentence pairs is utilized for the first-stage fine-tuning.

Real-time Interpretation Pipeline

Translation Pipeline

Speech-to-Text

Text-to-Text

Accent Removal

Operating Time Delay

Risk
[Warning]
Database
Basic Elements

Data Integration
[Data(Integration)]
New System
Connection Preparation

Summary
[Summary]
Details omitted...

Case Search
[Case(Law) Search]
Inquiry about Business Project
Related Laws

Gene
Discovery
[Genome Sequencing]
Bioinformatics
Data Analysis

Excitedly

Exciting Fun

Shout

Whisper

Keep things fresh

QUICK SETTING
User Manual
Simple Setup Only: 5-30 seconds

Please collect

I will definitely take it

Focus
Mindful Breathing

Glass

PET

Iron and Can Products

Available for Purchase After Login

Landline Phone Number

[OCR]

HIRO

[OCR]
[OCR]

[OCR]

10

B

reload

4/2/2023

Intel® Core™ Processor

HP Pavilion Laptop 15z

PHASE ONE

[OCR]

[OCR]

(2) E2E Real-time Streaming Voice Interpretation Model.
[Technical Architecture] InfiniSST + Whisper Large-v3 + LLM-based E2E Streaming Interpretation

Technical Element

Implementation Method and Content

Chunked Causal
Streaming ASR

Whisper Large-v3-based Chunk Level Causal Encoder. Real-time processing during speech.
Business jargon and product names specialized fine-tuning. Background noise filtering with VAD (Voice Activity Detection). WER target below 15%.

E2E Simultaneous Interpretation
Core Model

InfiniSST Framework: Wav2Vec2.0 Encoder + Adapter + Qwen2.5-7B LLM Decoder. Handles unlimited voice streams with sliding KV-Cache management.
Apply AlignAtt simultaneous interpretation policy. Target delay: StreamLAAL below 2 seconds.

Business Context
RAG Real-time Integration

Real-time referencing of corporate RAG knowledge DB during interpretation (initial prompt injection).
Prevents mistranslation of product names, model numbers, prices, and Incoterms. Domain term beam search weight applications.

Multilingual Support & Speaker
Separation

Korean <-> English, Chinese, Japanese, Spanish, German, French, Vietnamese, Arabic
(8+ languages). Automatic speaker differentiation with Speaker Diarization. Automatic language detection per speaker.

Platform Integration

Native plugins for Zoom, MS Teams, Google Meet. Mobile
app (iOS, Android). Low-latency streaming API for exhibitions and field use. Supports offline
on-premises deployment (sensitive information security).

(3) Export Document Automation Agent - 5 Core Operations

Document Type

AI Agent Processing Content and Enterprise Utilization

Multilingual
Product Pages &
Catalog

Korean product description -> Multilingual (English, Chinese, Japanese, Spanish, French) shopping mall optimized phrase automatic generation. SEO keyword and cultural localization. [Health Functional Foods, Cosmetics] Cafe24, Amazon API integration.

Quotation &
Quote

Reference unit price list with RAG -> Reflect trade standards such as Incoterms, payment terms, packaging units.
[Automotive Parts, Footwear] Automatic multilingual generation for OEM delivery quotations.

Auto Response for Overseas Inquiries

CRM, shopping mall API: Foreign language inquiry -> Intent classification -> RAG search -> Automatic multilingual response.
[Food] Automation of frequently asked questions related to ingredients, allergies, certifications.

Multilingual Contract Template

Supply contracts, NDA, MOU templates + enterprise terms injection -> Automatic multilingual draft creation.
[Automotive Parts] Global OEM supply contract multilingualization.

Exhibition, Seminar Real-time Interpretation Support

Voice interpretation + multilingual subtitles + Q&A chat translation integration. Zoom, Teams plugins.
[Footwear, Cosmetics] Real-time consultation support at overseas exhibitions (MIDO, Cosmoprof).

(4) Regulatory Certification Document Formatting Module [StunningBox Exclusive Differentiation]

Certification Type

Target Enterprise

Automation Content

eCTD (Electronic Common Technical Document)
FDA, EMA, PMDA, NMP
A

Functional Health Foods (FDA
Registration) Cosmetics (NMPA
China Registration)

ICH M4 5-module automatic formatting, Module 1
country-specific automatic generation, XML Backbone, PDF/A
automatic conversion.

CE Marking (EU Conformity Declaration)

Automotive Parts, Footwear
Cosmetics (EU Export)

EC Declaration of Conformity automatic formatting,
Technical Documentation translation and structuring.

FCC Certification (U.S. Radio Certification)

Electronics with electronic or wireless communication function and automotive parts

FCC Form 731 automatic formatting, technical document
English translation and structuring.

HACCP, Food Export
Documents

Food Manufacturers, Functional Health Foods

Automatic multilingual generation of ingredient tables, allergy labels, origin certificates,
automatic application of country-specific labeling standards.

3) Solution Development and Distribution Method to Target Enterprises

The AI Interpretation and Translation Solution of this project has already completed PoC in the global e-commerce field of genuine grip, proving commercialization viability.
Multilingual product page automatic generation and regulatory document translation are operated as a real service, and in this consortium, this verified AI core is mounted on the PCN SaaS platform, and expanded to all industries.

R&D Result Verification Method (4-Stage Framework)

Solution Development Method

Stage 1 Automatic Evaluation (Detailed)

Stage 2 Expert Evaluation (Quarterly)

Stage 1 Fine-tuning

Stage 2 RAG Integration

Stage 3 RLHF

3+ Billion Corpus
Extreme Beam LoRA

Vector DB+Hybrid Search
Exploration 70% ↓

Expert Check Vek
DPO Economic Learning

MQM 5-point scale human evaluation
once a quarter, total of 4 times

BLEU · WER · F1 · TEDS · StreamLAAL
CI/CD Pipeline Automatic Objective

Main AI Hype Curve (Domestic)

Main ScaleVex + Major Institutions

Solution ① E2E Interpretation

Solution ② Document & Regulation

Solution ③ Interpretation & Translation

Stage 4 Official Verification

Stage 3 Field Demonstration (3 times)

InfiniSST + Whisper

Location Information Module + CTD

NLL/EX▲ONE + RAG

Notification→Ray→Execution (5 o'clock A/B)

TTA/KAIC Official Test Certification

BLEU 65+

Natural as-is, Scent+Well-being

CE · FCC · HACCP

Precise measurement of smog duration and ratio

AI Performance+SW Quality+Stability

Water filter 40% ↑

Location Classification Zoom Practice

TEDS TDM-I

Encouragement: Demand-based Optimization

Performance from April to June in past years

[E2E Core] InfiniSST: Wav2Vec2+Adapter+Qwen2.5-7B · AlignAtt · Sliding KV-Cache

Performance Indicator Target Value

Delay 2 seconds ↓

BLEU 65↑

WER 15% ↓

F1 0.8+

TEDS 70%↑

SaaS Onboarding 5 Steps (Minimum 3 Days, No IT Staff Required)

Paper

Driving Age

Registration Recognition

Hourly Registration

Document Classification

STEP3

STEP1

STEP2

STEP4

STEP5

Development · Release

API Integration

Electronic Receipt

Insurance Selection Completed

SongBoWon

Work Day Reduced by 2 Hours+ | Cost Savings of 35 Million Won/Company Annually | 100%+ ROI within 12 Months

Voucher

Specification

Contingency: Redevelopment within 2 Weeks if Failed → Retest within 1 Month (Project Contingency at End)

Python 4 O'clock

50 sheets of spilling test paper

500 sheets+ of Sprinkled Items

1,000 sheets+ Globally

RND

KOTRA

Delivery

3.2 Billion Won

Sign-Up Rate 99.9% | API p95 500ms ↓ | SUS 68+ | User Satisfaction 4.0/5.0+

(1) Solution Development — 3-Stage Training Pipeline → Output of 3 Main Solutions

Classification

Core Technology

Development Content

Target Performance

3-Stage Training Based on NLLB/EXAONE
(Fine-tuning → RAG → RLHF). Utilizing 1.1 Million Corpus of Study Box. 3 Types of Industry-specific LoRA Adapter. Real-time Context Injection for Companies' RAG

Business Translation Model

BLEU 65+, Terminology Consistency
40%↑, Hallucinations
70%↓

Solution ①

E2E Real-Time Speech Interpretation

InfiniSST: Wav2Vec2+Adapter+Qwen2.5-7B
LLM. AlignAtt Simultaneous Interpretation Policy. Unlimited Stream with Sliding KV-Cache. Bidirectional in 8+ Languages. Speaker Separation, Zoom/Teams Integration

StreamLAAL 2s↓,
WER 15%↓, DER
15%↓

Solution ②

Five Types of Agents (Catalog · Quotation · Auto Response · Contract · Expo Interpretation). Regulatory Certification Formatting
(eCTD · CE · FCC · HACCP). 500,000 Corpus of Study Box for Proprietary Differentiation

Document Auto-Generation ·
Regulatory Certification

TEDS 70%+, Document Generation
Time 70%↓, F1 0.8+

Solution ③

(2) Distribution Method for Demand Companies

Required

Stage

Deliverables

Value from Company's Perspective

Execution Contents

SaaS Signup → Plan Selection
(Starter/Business/Enterprise)
→ Payment → Automatic Cloud Tenant Creation

1 Day

Enterprise-Specific SaaS Instance

No Separate Server/IT Staff Needed. Immediate Use Upon
Card Payment

1

Existing Terminology/Product Names
Reflected Instantly. 40%↑ Accuracy
than General AI
Industry-Specific Models Maximize
Specialized Term Accuracy
Seamless Integration with Existing
Business Systems. No Additional
Development Needed

Upload Company's Data (Product DB, Terminology Glossary, Price List) →
Automatic Parsing and Cleansing
→ RAG Vector DB Embedding

1~3 Days

Enterprise-Specific RAG Knowledge DB

Selection of Pre-Trained Models by Industry (Manufacturing/Retail/Food/IT)
→ Lightweight
LoRA Fine-Tuning → A/B Testing

1~2 Days

Custom Translation Model for Company

API Key Issuance → Integration with Shopping Mall
(Cafe24 · Amazon) · CRM · ERP
→ Test Execution

1 Day

API Integration Completed, Live Service Begins

Usage Logs → Quality Monitoring Dashboard
→ Monthly CI/CD Automatic
Retraining → Automatic Glossary Updates

Continuous

Monthly Quality Reports, Model
Autonomous Improvement

An Auto-Learning System
That Becomes More Accurate with Use

5

(3) Core Requirements and Reflections of Demand Companies
(3-1) Key Functions by Requirement
a. Structural Bottleneck in the Industry at Large

meaning

Item

numerical value

Problems faced by 3 demand companies are structural issues common to similar industries and scales. Overall structural issues common to such enterprises, the number of domestic exporting companies is a structural problem faced across the board. The industry-wide fixation on manual and outsourced structure for domestic export documents
Market size high cost issue
The language service itself is a large growth market, and automation of regulatory compliance is structurally present in market demand
The automation of regulatory responses grows at a rate 1.5 times that of the overall market growth rate—market is moving towards requiring automated solutions
Software market

approximately 93,000 companies (2024,
Korean Customs Service)

approximately 3.8 trillion KRW annually

53.9 billion USD

36.2 billion USD (CAGR
12.67%)

hundreds to tens of millions of KRW+
several months lead time
(Repeat occurrence by industry·country)

certification/document response costs per country

The cost and time of exports to multiple countries accumulates repeatedly by the number of countries × the number of products

Source: Korean Customs Service (2024), KTR Export Certification Support Center (2024), Element Korea Overseas Certification Cost Guide (2024), KOTRA Export Voucher Survey (2024), Mordor Intelligence (2025) summary based

b. Connection of Demand Companies' Respective AS-IS to Industry Statistics

Connection to Evidence Across the Industry

Individual AS-IS (Current)

Demand Companies

Even automotive parts and other manufacturing export companies repeatedly incur structural costs for country-specific certification and manual responses—Even large corporations (such as Kia Motors) experience similar bottlenecks by concurrently using in-house translation systems and external agencies.
Consumer goods export sectors like K-Beauty incur excessively high costs related to country-specific regulation and consultation compliance relative to sales volume—The high growth of the language service market (53.9 billion USD) itself evidences this demand
Special sectors such as highly regulated shoe industries experience hundreds to thousands of KRW per country and several months of lead time—A typical case of a 3.8 trillion KRW outsourcing market

Multilingual product manual translation/creation 72 hours

Sunlight Illucom
(Automotive Parts)

Global buyer email and
real-time consultation response average
24 hours

Gray Whale
(K-Beauty)

Country-specific certification/regulatory document
analysis/draft preparation 48 hours

Genuine Grip
(Special Footwear)

Requirements

Performance Indicators

Demand Companies

Core Technologies for Response

Consistency Requirement for Multilingual Catalog Terminology

Ontology-based Terminology Constraint Decoding

Sunlight Illucom specialized term F1
Processing Speed (2 seconds
within),
Document Structure Preservation Rate
Genuine Grip
(TEDS)

Minimize Delays in Real-time Buyer Consultation Response

StreamLAAL-Based Real-time Interpretation Pipeline

Requirement for Preservation of Tables/Format in Overseas Regulatory Documents

TEDS-Based Document Structure Preservation Agent

(3-2) Practical Use Plan and Advancement Plan

a. Practical use: Web Dashboard Construction

- API integration, choice of company-specific application method during linkage to demand company internal business systems

▽Binary -009 ▽Below -009

SEO Marketing Copy

Copy·Shorten

Open·Copy

Tag Input Window (Mac)

Title/Meta/Tags Auto-generation

+ Shopify·Amazon Compatible

6339 EA

Localization by Cultural Sphere

Light
radar fusion
Slip resistance (SRS)
Composite toe cap
Impact protection
Oil resisting

RAG setting

Hule

[OCR]

Terminology Glossary Fixation for Product Name·Dimension·Certification Name

[OCR]

/ Automatic Source Indication / Localization of Marketing Expression by Cultural Sphere

[OCR]

My Document Context in Footwear Export

SG-880 Product Specification Sheet, CE Type Examination Certificate, Export Unit Price·Quote, Footwear Export Terminology Standard Book,

Above Submitted Document List

Buyer Catalog·List

Genuine Grip Safety Shoe EC Declaration of Conformity

Export Declaration Number: PGS21-129

Construction Performance

Bucket List

Notice

Coupon Usage

Glossary

Product Deletion Estimate (0.4c)

Report

Domain

safety footwear+안전화, slip resistance(SRC)+미끄럼 방지, composite toe cap+복합 선심(편집 중), impact protection+충격 보호, CE marking+CE 마킹

Domain

safety footwear
slip resistance (SRC)
composite toe cap
impact protection
CE marking

Domain Profile

Domain

Footwear Safety Standards (CE) · EN ISO 20345

KEYSTONE SOLE PATTERN

Specification·Dimension Marks in Original Form (EN ISO 20345·S3·EU)

Size, Product Model Name (SC-880)

-Onboarding: Minimum 3 days per standard data baseline, typically real-use launch within 3 to 7 business days post data upload
- Advancement: Quarterly model update, automatic glossary reflection, history-based consultation
- Continuous Learning Pipeline Operation

b. Verification Plan for Demand Companies

Sunlight Illumi Co., Ltd.

Genuine Group Co., Ltd.

Gray Warming

Ignition·Extinction·Rotation

Light·Sound·Fisher

Path·Guide·Day·Path

Givenchy Lighting Parts Manufacturing

Special Industry Manufacturing

K-Beauty Division

Specialized Overseas Export Overtime

CE Marking · China NMPA Export

Audi/VW Tier1 Registration ('17)

Numerous Buyer Consultation Exhibitions

K-Beauty Global Marketing Regulatory Agency Self-Post-Numerous

Strategic Export by Type 3 (24)

Sound Product (25)

Global E-commerce PoC Completion

Provision of On-site Data

Provision of On-site Data

Provision of On-site Data

Product Catalog - Some Material Specifications

Supply Technical Test Properties · Quality Standard Sheet

CE-FDA Ingredient Document

K-Beauty Marketing Content

OEM Contract · Exhibition Script

Baker Wolseong Conversation

Export Customs Clearance · Shopping Mall Inquiries

Regulatory Agency Expenditure Documents

Audi/VW Remote Consultation Operation

Adaptive Translation and Interpretation Solution

Applied Translation and Interpretation Solution

Applied Translation and Interpretation Solution

Regulatory Certification Formatting

Technical Document Translation

Ensuring OEM Entry Term Consistency

Catalog Multilingualization · SEO Localization

Detailed Page Agent

CE FDA NMPA Ingredient Documentation Indigenization

E2E Real-time Interpretation

E2E Real-time Interpretation

Buyer Consultation · Exhibition On-site Interpretation

Exhibition On-site Interpretation Such as MIDO

Top Page Agent

Marketing Content Localization in 4 Languages

Contract Agent

Quotation · Inquiry Automation

Incoterms Quotation · Multilingual Template

eCTD Module

Functional Ingredient ICH M4 Conversion

OEM Supply Contract Multilingual Draft

(3-3) Field Application Effects for Demand Enterprises (AS-IS, TO-BE)

AS-IS

Key Improvement Indicators*

TO-BE (After Introduction, Interim Goal)

Demand Companies

Manual Translation Time
Sunshine Illucom
Reduced by More Than 50%
Consultation Response Lead Time
Gray Whale
Significantly Reduced by 97%
Regulatory Document Processing Time
Genuine Grip
Reduced by More Than 70%

Multilingual Product Manual Translation/Creation
-72 Hours

Within 24 Hours After Applying AI
(Proofreading Included)

Global Buyer Email·Real-time
Consultation Response Average - 24 Hours

Within 30 Minutes After Applying AI

Country-specific Certification·Regulatory Document Analysis/ Drafting - 48 Hours

Within 24 Hours After Applying AI

*Methodology for Calculating Key Improvement Indicators (Same as Announcement Performance Indicators Calculation Methodology)
① Comprehensive Survey of Demand Enterprises→ ② Exclude Outliers Based on Quartile Standard→ ③ Calculate Average (Determine AS-IS) → ④ Set Improvement Rate Goal (TO-BE)
Apply the same four-step procedure to all three demand enterprises to ensure the objectivity of AS-IS/TO-BE figures.

(4) Scheme for Securing Performance Verification System

Apply four-step verification (Automatic→Expert→On-site→Certified). Continuously
measure 5 KPI including BLEU 65+, WER 15%, and if underperform, redevelopment within 2 weeks. Verified in demonstration with three demand enterprises
that work time reduced by 2+ hours per day,
and cost savings of 35 million KRW per enterprise per year.
※ Key: Development→Dissemination→Verification are conducted simultaneously in an Agile manner and on-site feedback is immediately reflected in RLHF, implementing
SaaS that becomes more accurate with CI/CD automatic retraining as it is used more frequently.

1) Performance Indicators and Targets

Stage
Performance Indicator Name

First Year

Second Year

Third Year

Total

Weight (%)

Patent Application

1

1

1

3

25

Patent Registration

-

-

1

1

25

Expert Report

2

2

2

4

25

SW Registration

-

-

1

1

25

Total

3

3

5

9

100

※ The one patent registration in the third year is for the patent applications (3)

2) Performance Indicators and Measurement Methods

(1) Performance Indicators of Outputs

Proportion
in
total
Items

Classification

Evaluation Items
(Key Performance)

Unit

World-Class Level
(Holding Country/Institution)

Domestic Level Before R&D

R&D Target Value

Basis for Setting Targets

Performance Level

Performance Level

First Year

Second Year

Third Year

USA/Google Gemini
3.5 Live (70 languages
real-time), Germany/DeepL
However, Business
Domain·B2B Technical Terms
Not Supported

Model
Performance

Business Domain
Translation Accuracy
(BLEU)

score

20

Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.

-

60 or more

65 or more

Application of SacreBLEU, the standard metric for WMT.

Aiming to achieve world-class standards, applying ACL StreamAtt's StreamLAAL evaluation method.

Interpretation delay time (StreamLAAL)

sec

15

Delay of 2.5~3 seconds or more, such as by the USA/Google S2ST (with no Korean support)

Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.

-

2.5 seconds or less

2 seconds or less

Adoption of the Whisper's Common Voice·FLEURS WER/CER evaluation method.

Speech Recognition Error Rate (WER/CER)

%

15

For general domain by USA/OpenAI Whisper Large-v3, WER is 10~12%

Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.

-

20% or less

15% or less

Application of the TEDS standard metric of PubTabNet.

Document Structure Preservation Rate (TEDS)

%

10

-

-

-

60% or more

70% or more

Remains at the level of static registration features like the Papago Plus glossary function, lacking automatic extraction and real-time search augmentation systems.

Term Extraction (F1)

score

10

Based on common glossary like UK/RWS Holdings SDL Trados, F1 level is 0.6~0.7

-

0.7 or more

0.8 or more

Precision and Recall evaluation based on approved Gold Standard glossaries from demand companies.

Proportion
in
total
Items

Classification

Evaluation Items
(Key Performance)

Unit

World-Class Level
(Holding Country/Institution)

Domestic Level Before R&D

R&D Target Value

Basis for Setting Targets

Performance Level

Performance Level

First Year

Second Year

Third Year

Reduced the average working time from 72 hours to 18 hours.

Field Application (Effectiveness before and after solution adoption)

Haeppit Illucom

Product manual translation

Time

5

-

72

-

60

54

Reduced daily work time for trade personnel from 4 hours to 2 hours

Understanding and translating the context of entire email threads

Time

5

-

4

-

1,5

2

Translation of regulatory certification documents

Reduced business costs by 50%

ten thousand won

5

-

1,000

-

600

500

Due to lack of domain-specific global translation, created pages in three languages (English, Chinese, Japanese)

Gray Whale

Global shopping mall page

Cases

5

-

-

-

2

3

Real-time buyer consultation interpretation

Achieved interpretation delay time of 2 seconds

Sec

5

-

6

-

2.5

2

Reduced daily work time for trade personnel from 4 hours to 2 hours

Understanding and translating the context of entire email threads

Time

5

-

4

-

1,5

2

Translation of regulatory certification documents

Reduced business costs by 50%

Genuine Grips

ten thousand won

5

-

300

-

180

150

Reduced the average working time from 72 hours to 18 hours.

Product manual translation

Time

5

-

72

-

60

54

Reduced daily work time for trade personnel from 4 hours to 2 hours

Understanding and translating the context of entire email threads

Time

5

-

4

-

1,5

2

※ The domestic level before the field application of R&D corresponds to the level before the solution adoption by the demand companies.

(2) Evaluation Methods and Environment

Evaluation Items (Performance Indicators)

Evaluation Environment (Measurement Method)

Evaluation Method

Evaluation Content: Compared FLORES-200 and regulatory document test set (1,000 sentences) with professional translations · Evaluation Procedure: Compared AI translation results and correct translations using SacreBLEU · Formula: SacreBLEU(Test, Reference) · Variable Explanation: Test=AI translation, Reference=Professional Correct Translation

Business Domain Translation Accuracy (BLEU)

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Evaluation Content: Performed 100 sentences real-time interpretation of Korean↔English · Evaluation Procedure: Measured average delay time from speech initiation to translation output · Formula: Σ(g(t)-t) / N · Variable Explanation: g(t)=AI translation output time, t=User speech initiation time, N=Total number of sentences

Interpretation Delay Time (StreamLAAL)

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Evaluation Content: Compared 10 hours of conference speech with correct scripts · Evaluation Procedure: Calculated error rate by substitution, deletion, and insertion errors · Formula: (S+D+I) / N ×100 · Variable Explanation: S=Substitution, D=Deletion, I=Insertion, N=Total number of words

Speech Recognition Error Rate (WER/CER)

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Document Structure Preservation Rate (TEDS)

Evaluation Content: Converted 100 PDF files to HTML and compared with the original · Evaluation Procedure: Automatically calculated document structure (Tree) similarity · Formula: TEDS (Tree Edit Distance Similarity) · Variable Explanation: Structural similarity between the original HTML and the conversion result (0~1)

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Term Extraction (F1)

Evaluation Content: Classification evaluation of regulatory document test set (3,000 sentences) · Evaluation Procedure: Calculated Precision and Recall, then computed F1 · Formula: 2×P×R / (P+R) · Variable Explanation: P=Precision, R=Recall

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Effect of reducing working time in product manual translation tasks using AI · Evaluation Content: Compared translation times before and after AI introduction for 5 types of product manuals (10 cases total) · Evaluation Procedure: Measured average working time by translating the same document using traditional and AI methods · Formula: Average time reduction = (Working time before AI introduction − Working time after AI introduction) / Sample count

Product Manual Translation

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Effect of understanding the context and reducing working time in email thread translation tasks · Evaluation Content: Compared business hours before and after AI introduction for 20 buyer email threads · Evaluation Procedure: Measured average daily working time by applying traditional and AI methods by the same personnel · Formula: Average daily time reduction = Business time before AI introduction − Business time after AI introduction

Understanding and Translating Email Thread Context

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Cost reduction effect of translating regulatory and certification documents · Evaluation Content: Compared translation costs before and after AI application for 10 regulatory and certification documents · Evaluation Procedure: Assessed and compared translation costs of the same document before and after AI introduction · Formula: Cost reduction rate = (Cost before introduction − Cost after introduction) / Cost before introduction × 100

Translation of Regulatory Certification Documents

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Creation of Global Shopping Mall Pages

Definition: Range of support for automatic generation of multilingual pages for global shopping malls · Evaluation Content: Created multilingual product pages for 10 representative products · Evaluation

Procedure: Reviewed language-specific results to verify the number of supported languages · Formula: Number of supported languages = Number of generated languages meeting quality standards

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Real-time interpretation response delay time during buyer consultation · Evaluation Content: Measured average response time for 100 Korean-English consultation speeches · Evaluation Procedure: Automatically measured time from speech end to translation output · Formula: Average delay time = Σ(Translation output time − Speech end time) / Number of speeches

Real-time Buyer Consultation Interpretation

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Effect of understanding the context and reducing working time in email thread translation tasks · Evaluation Content: Compared business hours before and after AI introduction for 20 buyer email threads · Evaluation Procedure: Measured average daily working time by applying traditional and AI methods by the same personnel · Formula: Average daily time reduction = Business time before AI introduction − Business time after AI introduction

Understanding and Translating Email Thread Context

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

Definition: Cost reduction effect of translating regulatory and certification documents · Evaluation Content: Compared translation costs before and after AI application for 10 regulatory and certification documents · Evaluation Procedure: Assessed and compared translation costs of the same document before and after AI introduction · Formula: Cost reduction rate = (Cost before introduction − Cost after introduction) / Cost before introduction × 100

Translation of Regulatory Certification Documents

Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

(5) Quantitative Comparison Indicators— Analysis of Numerical Gaps In addition to qualitative coverage comparison, a quantitative comparison based on actual disclosed figures is presented. Figures are based on each company's public data and announcements, and the platform's target figures remain consistent with the previously presented performance indicators comparison table (items 5, 6, 12, 20 combined).

Comparison Items

Unit

Competitors/General Standards

numerical value

Platform Target

Translation Quality (Expert Preference)

Multiple

GPT-4 (vs DeepL)

1.7 times lower preference compared to DeepL (DeepL's own blind test)

Aim to achieve equivalent or superior levels with domain-specific fine-tuning (to be verified with internal A/B benchmark)

Translation Post-Editing Requirement

Multiple

GPT-4 (vs DeepL)

Requires 3 times more editing compared to DeepL (based on DeepL's own announcement)

Aim to minimize editing needs with specialized terminology consistency metric (over 85% term matching rate)

Real-time Interpretation Delay

Sec

Transync AI·Tiro
etc. (General-purpose Conversation)

Less than 0.5 seconds

Within 2 seconds (includes conditions for consistency of technical terms + RAG search + document linkage)

Document Preparation Lead Time (Human-based)

Duration

TransPerfect·RWS
etc.

Weekly Unit

Target for Demand Companies: Within 2 hours (up to 97% reduction)

Translation Rate (Domestic Professional Agency)

KRW/Page

Domestic Professional Translation Agency (based on 550 characters)

Typically 20,000 to 50,000 KRW, up to 150% surcharge for specialized fields (contracts, etc.)

Significant reduction through AI automation (exact savings rate to be confirmed after internal cost estimation)

AI Solution Construction Costs

KRW

Hyperscience etc. IDP Solutions

Hundreds of millions to billions of KRW

No initial construction costs due to SaaS subscription model

Regulatory Certification Document Response Costs (Per Country)

KRW·Duration

Average for Domestic Export Companies (StunningBiz AI Survey Data)

Tens to hundreds of millions of KRW + several months lead time

Goal to significantly reduce through automation (verified through demand company demonstration)

※ The 'Real-time Interpretation Delay' item clearly states that a competitor leads in sheer speed—however, the competitor's product is solely for casual conversation and lacks features for consistency of technical terms, RAG search, and document linkage. It's important to specify in the notes that this platform aims for under 2 seconds, despite including these features. Translation rates, construction costs, and regulatory response costs are areas currently lacking precise reduction figures, pending internal cost assessments and verification through demand company demonstration.

3. Preceding Research and Development

1) Preceding R&D History

(1) Leading Organization PCN Co., LTD's Preceding Patents and Technologies

Task Name/Patent Name

Core Achievements/Content

Organization·Duration

Remarks

Ontology-based Search Service System and Method

Implementation of a semantic-based search engine. Technology to improve search accuracy through term relationship inference

PCN Co., LTD March 2019
Registered No. 10-1958811

Patent Registered

Intelligent Search System and Method Based on RDF Triple Data

Structured data-based intelligent search. Preceding technology for RAG vector search

PCN Co., LTD April 2019
Registered No. 10-1972127

Patent Registered

Open API-based Content Service Interface Provision System and Method (OASIS)

API auto-generation platform. Secured architecture for multiple API integration and linkage

PCN Co., LTD August 2013
Registered No. 10-1296321

Patent Registered
GS Certification

Oz-i AI Platform - STT-based Automatic Meeting Summary - Multilingual Translation Service

Integrated platform applicable to industrial sites, metaverse, and robots with AI. Completed implementation of STT speech recognition + multilingual translation + sLLM. Applied to the aviation industrial complex metaverse project

PCN Co., LTD
2024~Present

Commercial Platform
Multilingual Translation
PoC

Supreme Court Criminal Electronic Litigation System Construction (110 billion scale)

Verified capabilities in large-scale platform construction and operation. Capability in public SI business execution

PCN Co., LTD
2023
Contract Awarded

Public SI
Large Reference

(2) Collaborating Organization Stunning Box Co., LTD and Subcontractor's Preceding Technologies

Task Name/Patent Name

Core Achievements

Organization·Duration

Remarks

RAG (Search-Augmented Generation) Solution Utilizing Generative AI

Patent application and progress for <RAG (Search-Augmented Generation) Solution Utilizing Generative AI> by Kumoh National Institute of Technology Industry-University Cooperation Group (Inventor: Professor Seong-ryeol Kim)

Kumoh National Institute of Technology ->
Stunning Box Co., LTD

Patent Application
10-2024-0147669

Clinical·Regulatory Professional Translation Multi-Agent Prototype

8-types Multi-Agent Translation Pipeline PoC, regulatory document RAG verification, document structure preservation rendering

Stunning Box Co., LTD
2024~Present

B2B PoC Completed

2) Plan for Utilization of Preceding R&D Results in the Execution of This Project

Preceding R&D
Outcomes

Period·Cost Reduction Through Utilization

Organization

Specific Utilization Method for This Project

Approximately 6 months reduction compared to new platform development. Approx. 50 million KRW reduction in cloud infrastructure construction costs

Directly utilized as the base infrastructure of the SaaS platform for this project. Combined Oz-i's STT engine with Whisper Large-v3 as a preprocessing module for E2E voice interpretation. Directly leveraging sLLM serving architecture (vLLM, TensorRT-LLM) for deployment of translation and interpretation models.

Oz-i AI Platform (STT·Multilingual Translation·sLLM)

PCN

Ontology Search
Patent (Semantic
-based Search)

Applied to the core design of the RAG vector search engine. Implemented the ontology-based semantic similarity algorithm on ChromaDB/Pinecone with a hybrid search layer (BM25+Dense Retrieval). Directly contributes to improving semantic search accuracy for corporate terms and product names.

Approx. 3 months reduction in RAG search engine design. No separate search algorithm research needed

PCN

Reused RESTful API gateway architecture for integration with demand companies, shopping malls, and CRMs. Applied OASIS's API auto-generation, documentation, and version management functions to the API

B2B2B service of this project. Basis for API integration designs of Cafe24, Amazon, Naver Shopping Mall.

OASIS API
Platform (GS
Certification)

Approx. 4 months reduction in API integration module development. Approx. 30 million KRW reduction in API standardization and documentation costs

PCN

Directly applied know-how in designing and operating multitenant SaaS architecture. Embedded operating systems such as Kubernetes auto-scaling, CI/CD pipeline, monitoring (Prometheus+Grafana) into this project's SaaS. Technical basis for ensuring 99.9% SLA availability.

Cloud SaaS
Operational
Capability (25
Years Record)

Approx. 4 months reduction in SaaS operational system setup. Approx. 20 million KRW savings in stability verification costs

PCN

Directly incorporated into the core patent technology of this project's RAG architecture. Applied on-device secure search technology to protect corporate confidential data (price lists, contract terms). Provides theoretical basis for the Evidence Grounding method to prevent LLM hallucination.

StunningBox

RAG Solution
Patent

RAG core design approximately
3 months reduction. Security
search separately developed not
required

Directly utilized as a basis for the development of '5 export document automation agents' in this project from the 8 types of Multi-Agent pipelines. Applied the eCTD document structure preservation technology verified in the PoC
to the regulatory certification formatting module. Reflected in the strategy to

achieve professional terminology extraction F1 0.8+ with the RAG validation results of clinical terms

StunningBox

Multi-Agent
Translation
PoC (Document
Structure Preservation)

Agent Prototype
approximately 4 months
reduced. Regulatory document
structure analysis costs reduced
by approximately 40 million KRW

Immediately deployed as core data for AI model training in this project. 50 million sentence pairs of regulatory corpus → trained regulatory certification formatting module. 60 million sentence pairs of business corpus → fine-tuning of translation models + RAG glossary. Possibility to commence model training from the first year without separate data collection and cleansing

StunningBox

1.1 million+
professional translation
corpus (Accumulated over 15
years)

Data collection and cleansing
approximately 8 months reduced.
Data construction costs reduced
by approximately 200 million KRW
(maximum differentiation)

4. R&D and Commercialization Capability

1) Plan and Capacity for Team Formation

(1) Structure and Role Allocation of the Consortium

a. Why this Consortium — The Inevitability of the Combination of PCN × StunningBox

The project requires 'AI platform technology' and 'professional translation and interpretation domain knowledge' simultaneously, which cannot be achieved by one or the other alone.

The combination of PCN (Platform·SaaS·API) and StunningBox (AI Translation Model·1.1 million corpus·Regulatory Certification) is the only combination that enables achievement of goals within 24 months.

The principal R&D institution has experience in numerous AI R&D projects, possessing its own AI ontology platform, and plays a role not only in mere platform implementation but also in optimizing and applying AI model outcomes to actual services.

Particularly, by combining ontology-based knowledge structure with AI models to implement domain-specific inference functions and agent-based automation functions, the practical utility value of the AI technology is maximized.

Classification

PCN [Lead] — Technology·Platform

StunningBox [Joint-1] — AI Model·Data

Possession of a 15-year professional translation corporate dataset. Exclusively holds a business and regulatory parallel corpus of over 1.1 million sentence pairs. RAG patent registration. Completion of Multi-Agent Translation PoC. eCTD·CE·FCC regulatory document structuring technology

Fine-tuning of translation and interpretation AI models (NLLB/EXAONE), E2E speech interpretation model (InfiniSST), RAG architecture design, regulatory certification formatting module, data cleansing and revision

25-year IT/AI company (Revenue 27 billion KRW, 170 employees). SaaS platform·API integration·Cloud infrastructure construction·operation

Core Competencies

SaaS platform development and operation,
API integration for malls·CRM·ERP, customer
onboarding, cloud infrastructure (AWS),
CI/CD pipeline, commercialization·marketing
overall management

Handling in This Project

Installed StunningBox's specialized AI models on
PCN's proven platform infrastructure →
Development period reduced by 32 months, costs
reduced by 340 million KRW

The model trained on StunningBox's 1.1 million corpus is
distributed to all industries via PCN SaaS platform →
Data construction 8 months reduced

Synergy upon
Combination

(2) Commercialization Organization Composition Chart

Organization

Participating Personnel

Assigned Technology Details

PCN SaaS platform development and integration, API integration (malls·CRM), cloud
[Lead] infrastructure construction, customer onboarding, commercialization overall management
StunningBox AI translation and interpretation model fine-tuning, E2E speech interpretation, RAG architecture,
[Joint-1] regulatory certification formatting, data cleansing and revision
Customer
3 companies provide on-site data by industry, solution proof and verification, effect measurement
[Joint-2] cooperation, feedback provided

Total responsibility except 10 persons
(Total 11 persons)

Co-responsible person except 6 persons
(Total 7 persons)

2 persons per company responsible
(Total 6 persons)

2) Researcher Status (Lead Organization - PCN

Name

Affiliation·Position

Education·Major

Role during Project Execution

PCN
Director

Research Lead. Overall platform development responsibility, commercialization strategy establishment, consortium coordination

Um Sang-hyun

Master/AI·SW

PCN
Director

Choi Beom-jin

Bachelor/SW Engineering

Establishment and advancement of domain knowledge-based ontology

PCN
Manager

SaaS platform architecture design, cloud infrastructure construction and operation

Lee Kyung-soo

Bachelor/Backend

PCN
Assistant Manager

Bachelor/AI·MLOps

AI model deployment and serving (vLLM), CI/CD automation, performance monitoring

Kim Jung-woo

PCN
Manager

Design and implement application AI logic (agent, automation functions, etc.)

Choi Seung-bum

Master/AI·SW

PCN
Chief

Customer onboarding automation, dashboard development, user training and customer support

Lee Hye-jung

Master/AI·SW

PCN
Manager

Data collection and cleansing for customers, RAG vector DB construction, onboarding support

Lee Jae-man

Bachelor/Data

PCN
Chief

SaaS platform architecture design, cloud infrastructure construction and operation

Hong Hyo-jin

Bachelor/Backend

PCN
Team Leader

System testing, quality assurance, TTA/KAIC official verification response

Kim Tae-heon

Bachelor/QA

PCN
Assistant

Design and implement application AI logic (agent, automation functions, etc.)

Park Seo-ri

Bachelor/AI

PCN
Assistant Manager

Seo Min-gyu

Bachelor/Security

Cloud security, personal information anonymization, SLA monitoring

PCN
Researcher

Data collection and cleansing for customers, RAG vector DB construction, onboarding support

New Hire ①

Bachelor/Data

PCN
Researcher

New Hire ②

Bachelor/Security

Cloud security, personal information anonymization, SLA monitoring

PCN
Researcher

System testing, quality assurance, TTA/KAIC official verification response

New Hire ③

Bachelor/QA

PCN
Researcher

Design and implement application AI logic (agent, automation functions, etc.)

New Hire ④

Bachelor/AI

PCN
Researcher

Customer onboarding automation, dashboard development, user training and customer support

New Hire ⑤

Bachelor/SW

PCN
Researcher

Commercialization and marketing execution, KOTRA·Trade Association linkage,
customer management

New Hire ⑥

Bachelor/Marketing

3) Researcher Status of Participating Institutions - StunningBox

Affiliation·Position

Education·Major

Name

Role during Project Execution

Stunning Box Co., Ltd.
Manager

Bachelor's/Expert in Data
15 years

Co-Principal Investigator. Quality supervision and construction of 1.1 million corpus
Overall supervisor, Co-inventor of RAG Patent

Jeong Sumin

Stunning Box Co., Ltd.
Manager

Bachelor's/Developer
15 years

AI Model Fine-tuning Design, E2E Speech Interpretation Model Architecture, RAG System Design

Eom Junhyeon

Master's/Planner
5 years

Quality management of regulatory documents (eCTD·CE·FCC) and planning the overall flow of solutions

Yun Inhee

Stunning Box Co., Ltd.

Stunning Box Co., Ltd.
Manager

AI Model Quality Verification, Translation Result Evaluation (BLEU·MQM), Field Verification Support and Feedback Collection from Demand Companies

Kang Hyojeong

Bachelor's/Development

Stunning Box Co., Ltd.
Manager

Bachelor's/Translation·Data

Dataset Construction·Labeling, Standardization of Technical Terminology Dictionaries,
eCTD XML Structuring·PDF/A Conversion

Ma Jaemin

Stunning Box Co., Ltd.
Manager

Bachelor's/Regulatory Data Supervision

LLM Fine-tuning·RLHF Training /
Quality Management of Regulatory Documents (eCTD·CE·FCC)

Lee Junu

Stunning Box Co., Ltd.
Manager

Bachelor's/UX Design

Jo Min

UI/UX Design

Master's/Translation Review
and Expert Usability Inspection

New Hire ①

Stunning Box Co., Ltd.
Researcher

Dataset Construction·Labeling, Standardization of Technical Terminology Dictionaries,
eCTD XML Structuring·PDF/A Conversion

Stunning Box Co., Ltd.
Researcher

LLM Fine-tuning·RLHF Training Practice, E2E Speech Interpretation Model Training, Model Performance Evaluation·Optimization

New Hire ②

Master's/AI·NLP

Stunning Box Co., Ltd.
Researcher

Bachelor's/Data
Engineering

Corpus Refinement Pipeline Construction, RAG Vector DB Embedding·Indexing, Data Quality Verification (IAA 0.85+)

New Hire ③

Master's/Translation Review
and Expert Usability Inspection

Development of Regulatory Certification Formatting Module
(eCTD·CE·FCC·HACCP), Document Auto-generation
Agent Backend Development

New Hire ④

Stunning Box Co., Ltd.
Researcher

4) Patents, Papers, Awards Related to This Project

(1) Three Demand Companies — Demonstration Verification Participation Capability

Export Performance·Capability

Industry

Contents of Demonstration Participation

Demand Companies

Contribution to This Project (Data·Verification)

Provision of Technical Specification·Quality Criteria·OEM Contract·Exhibition Script. Quantitative Measurement of Time·Cost Reduction

Audi/VW Tier1 Registration
(2017). $30 million Export
Certification (2024). Ulsan New
Factory (2025)

German·English·Japanese
Translation Verification. Buyer
Consultation Interpretation
Verification

Automotive
Lighting Parts

Sunlight Ilucom Co., Ltd.

Verification of English·Chinese·Vietnamese
Translation. Multilingual Catalog

Special Shoes Manufacturing

Expert in Export and Customs Clearance.
Numerous Buyer Consultations·Exhibition Participations

Provision of Buyer Negotiation Dialogues·Product Catalog
Specifications·Raw Material Manuals. Exhibition Interpretation On-site
Verification

Genuine Grip Co., Ltd.

Provision of CE·FDA Component Documents·Marketing Content·Regulatory
Submission Documents. Measurement of Automated Certification Document
Effectiveness

Verification of English·Chinese·Japanese·
French Translation. Regulatory Certification Document Verification

K-Beauty Cosmetics

CE Marking·China NMPA
Export. K-Beauty Global Marketing

Gray Whale

Three PCN Patents (Ontology Search Patent, OASIS GS Certification, Oz-i Platform) + Stunning Box Two Patents (RAG Patent, 1.1 million Corpus) — Refer to Chapter 3, Previous Research and Development for Details
※ This consortium is the only combination that embodies the 'Technology (PCN Platform) × Data (Stunning Box 1.1 million Corpus) × Market (4 Demand Industries)', which is the key foundation to achieving TRL 4→7 and commercialization within 24 months.

5. Technology Development Schedule

1) Overall Development Schedule (2026.7 ~ 2028.6, Total 24 months)

■ Stunning Box Responsibility

■ PCN Responsibility

■ Joint Execution

■ Demand Company Participation

11
~1
2

No
.

Development Content

Responsibility

7~8

9~
10

1~2

3~
4

5~
6

7~
8

9~1
0

11~1
2

1~2

3~4

5~6

First Year
(2026.
7~202
7.6)

Second Year
(2027.7
~2028.6
)

Requirements Definition
·Architecture Design

1

Joint

●

●

Business·Regulatory Corpus Collection
·Refinement ·Term
DB

2

Stunning

●

●

●

Field Data Collection from Three Demand Companies
·Anonymization

3

Joint

●

●

●

●

Business Translation
Model Fine-tuning
(NLLB/EXAONE)

4

Stunning

●

●

●

●

RAG Corporate Knowledge DB ·
Hybrid Search
Engine

5

Stunning

●

●

●

6

E2E Streaming
Speech Interpretation Model (InfiniSST)

Stunning

●

●

●

●

●

SaaS Platform
·Cloud Infrastructure Construction

7

PCN

●

●

●

●

●

●

API Integration
(Cafe24·Amazon
·CRM·ERP)

8

PCN

●

●

●

●

Five Types of Export Document
Auto-generation Agents

9

Joint

●

●

●

●

●

Regulatory Certification Document
Formatting Module
(eCTD·CE·FCC)

10

Stunning

●

●

●

●

●

11

Web·Mobile
UI/UX ·

PCN

●

●

●

●

Zoom/Teams
Plugin

RLHF Reinforcement Learning·Model
Enhancement

12

Stunning

●

●

●

Alpha Test
(Client Company 3 Firms
Internal)

13

Joint

●

●

Beta Test
(Client Company 3 Firms
Practical Application)

14

Joint

●

●

●

CI/CD Automation
· Monitoring Da
shboard

15

PCN

●

●

●

Empirical Validation (3
Firms Full · A/B
Comparison · Performance
Measurement)

16

Joint

●

●

TTA/KAIC Official
Verification · Co
mmercialization Preparation

17

Joint

●

●

2) Detailed Role Distribution by Institution

Responsibility

Deliverables

Development Items

Detailed Execution Content

Oz-i Based Multi-Tenant SaaS Architecture Construction. AWS Cloud Infrastructure (EKS,

RDS, S3). 99.9% Availability SLA Operating
System

PCN

SaaS Platform · Cloud
Infrastructure

SaaS Platform
v1.0, Cloud In
frastructure Operation Manual

API Gateway,
Shopping Mall Integration SDK,
API Documentation

OASIS Architecture Based RESTful API
Gateway. Cafe24·Amazon·Naver
Shopping Mall Integration. CRM·ERP Integration Module

PCN

API Integration Module

Web App v1.0, Mobile
App v1.0, Zoom Pl
ugin

React Based Dashboard. Client Company Onboarding UI. Zoom·MS Teams
Native Pluggin. Mobile App (iOS·Android)

PCN

Web·Mobile UI/UX · Plugin

Kubernetes Auto Scaling. CI/CD Pipeline (GitHub Actions).
Prometheus+Grafana Monitoring. Auto
Relearning Pipeline

PCN

CI/CD · MLOps · Monitoring

CI/CD Pipeline,
Monitoring Dashboard,
SLA Report

1.1 million+ sentence pairs cleansing·de-identification·labeling. Industry-specific dictionary construction. Data quality validation (IAA 0.85+)

Refined Corpus 1.1
Million+, 3
Types of Terminology Dictionary, Quality Report

StunningBox

Corpus Collection · Refinement · Terminology DB

NLLB/EXAONE Domain Fine-Tuning
(LoRA). Three industry-specific adapters. DPO-based RLHF. Translation Expert Feedback Loop

Translation Model
v1.0~v3.0, 3 Industry-
Specific Adapters

StunningBox

Translation Model Fine Tuning ·
RLHF

InfiniSST:
Whisper+Wav2Vec2+Qwen2.5-7B.
AlignAtt Policy. Sliding KV-Cache. 8
+ Languages Bidirectional

StunningBox

E2E Speech Interpretation Model

E2E Interpretation Model
v1.0, Streaming
API

RAG Patent-based Hybrid Search Engine.
eCTD·CE·FCC·HACCP Automatic Formatting.
XML/PDF/A Auto Conversion

RAG Engine v1.0,
Regulatory Certification Formatting
Module v1.0

StunningBox

RAG Architecture · Regulatory Certification
Formatting

Alpha→Beta→Empirical 3-phase Verification. A/B Comparison Measurement. Work Hours·Cost Reduction Quantitative Data Collection

Alpha·Beta Test Report, Empirical Performance Report

Joint

Client Company Empirical · Validation

Official Agency Test Request→Execution→Certificate Issuance. AI Performance+SW Quality+Service Stability Comprehensive Verification. Commercialization SaaS Launch Preparation

Joint

TTA/KAIC Official Verification · Co mmercialization

Official Certification, Commercialization SaaS v1.0, Business Plan

3) Key Milestones by Year

Milestone

Timeline

Achievement Criteria (Go/No-Go)

First Year BLEU 55+, WER 20%↓, Interpretation Delay 3 secs ('26.12) ↓. If Not Met, Data Augmentation and Reassessment
First Year BLEU 60+, WER 17%↓, Alpha Functionality Norm ('27.6) Rate 90%+. Go: Beta Entry
Second Year BLEU 63+, StreamLAAL 2.5 secs↓, Satisfaction ('27.12) 3.5/5.0+. Go: Empirical Entry

Corpus Refinement Completed, Translation Model v1.0, E2E Prototype, SaaS Infrastructure Construction

Alpha Test Completed, API Integration Completed, Document Agents 3 Types, Model v2.0

Beta Test Completed, E2E 2 seconds Achieved, Regulatory Certification Module v1.0, Agents 5 Types Completed

BLEU 65+, WER 15%↓, StreamLAAL 2 Seconds↓, F1 0.8+, TEDS 70%+. Work Day Reduced by 2 Hours+

Empirical Validation Completed, TTA/KAIC Official Certification, SaaS v1.0 Commercialization Launch

Second Year ('28.06)

6. Research Expense Use Plan, etc.

1. Research and Development Cost Use Plan

1) Research and Development Cost Support and Burden Plan

(Unit: Thousand Won)

Government-Supported Research and Development Cost

Institution-Burden Research and Development Cost

Classification

Total

Year

Institution Role 1」

Research and Development Institution Name

Cash

Cash

In-kind

Subtotal

Cash

In-kind

Total

1

Principal

PCN Co., Ltd.

312,500

10,425

93,825

104,250

322,925

416,750

1

Participant

StunningBox Co., Ltd.

312,500

10,425

93,825

104,250

322,925

416,750

2

Principal

PCN Co., Ltd.

625,000

20,850

187,650

208,500

645,850

833,500

2

Participant

StunningBox Co., Ltd.

625,000

20,850

187,650

208,500

645,850

833,500

3

Principal

PCN Co., Ltd.

312,500

10,425

93,825

104,250

322,925

416,750

3

Participant

StunningBox Co., Ltd.

312,500

10,425

93,825

104,250

322,925

416,750

Grand Total

2,500,000

83,400

750,600

834,000

2,583,400

3,334,000

2) Research and Development Cost Use Plan

(1) Research and Development Cost Use Plan by Institution

(Unit: Thousand Won)

Research and Development Cost

Research and Development Institution Name

Research
Development
Cost Other
Support
Fund5」

Research
Inducement
Calculation
Standard
Amount6」

Direct Cost

Indirect
Cost

Total

Student
Labor Cost

Research Facility and
Equipment Cost

Labor Cost

Research
Material
Cost

Commissioned
Research
Development
Cost

International Joint Research and Development Expenses

Research and Development Burden Expenses

Research Activity Expenses

Research Allowance

Subtotal

General 1」

Special Case 2」

General 3」

Special Case 4」

355,3
70

1,271,
700

20,00
0

PCN Co., Ltd.

Cash

916,330

1,291,700

375,3
00

In-kind

375,300

375,300

1,291,63
0

355,3
70

1,647,
000

20,00
0

Subtotal

1,667,000

406,7
00

1,245,7
00

46,00
0

Corporation Stunning Box

Cash

784,000

55,000

1,291,700

375,30
0

In-kind

375,300

375,300

406,7
00

1,621,0
00

46,00
0

Subtotal

1,159,300

55,000

1,667,000

762,0
70

2,517,4
00

66,00
0

Grand Total

Cash

1,700,330

55,000

2,583,400

750,60
0

In-kind

750,600

750,600

762,0
70

3,268,0
00

66,00
0

Total

2,450,930

55,000

3,334,000

(2) Annual Usage Plan

(Unit: Thousand Won)

Research and Development Cost

Year

Research
Development
Cost Other
Support
Fund5」

Research
Inducement
Calculation
Standard
Amount6」

Direct Cost

Indirect
Cost

Total

Research Facilities and Equipment Expenses

Labor Cost

Student Labor Costs

Research
Material
Cost

Commissioned
Research
Development
Cost

International Joint Research and Development Expenses

Research and Development Burden Expenses

Research Activity Expenses

Research Allowance

Subtotal

General

Special Case

General

Special Case

20,00
0

208,5
20

633,8
50

1

Cash

405,330

12,000

645,850

187,6
50

In-kind

187,650

187,650

20,00
0

208,5
20

821,5
00

Subtotal

592,980

12,000

833,500

25,00
0

370,7
00

1,258,
700

2

Cash

863,000

33,000

1,291,700

375,3
00

In-kind

375,300

375,300

1,238,30
0

25,00
0

370,7
00

1,634,
000

Subtotal

33,000

1,667,000

10,00
0

182,8
50

624,8
50

3

Cash

432,000

21,000

645,850

187,6
50

In-kind

187,650

187,650

10,00
0

182,8
50

812,5
00

Subtotal

619,650

21,000

833,500

1,700,33
0

55,00
0

762,0
70

2,517,
400

Grand Total

Cash

66,000

2,583,400

750,6
00

In-kind

750,600

750,600

2,450,93
0

55,00
0

762,0
70

3,268,
000

Total

66,000

3,334,000

(3) Annual Usage Plan by Research and Development Institution

A. Name of Lead Research and Development Institution: PCN Co., Ltd.

(Unit: Thousand Won)

Research and Development Cost

Year

Research
Development
Cost Other
Support
Fund5」

Research
Inducement
Calculation
Standard
Amount6」

Direct Cost

Indirect
Cost

Total

Research Facilities and Equipment Expenses

Labor Cost

Student Labor Costs

Research
Material
Cost

Commissioned
Research
Development
Cost

International Joint Research and Development Expenses

Research and Development Burden Expenses

Research Activity Expenses

Research Allowance

Subtotal

General

Special Case

General

Special Case

109,
095

316,
925

1

Cash

207,830

6,000

322,925

93,8
25

In-kind

93,825

93,825

109,
095

410,
750

Subtotal

301,655

6,000

416,750

163,
850

637,
850

2

Cash

474,000

8,000

645,850

187,
650

In-kind

187,650

187,650

163,
850

825,
500

Subtotal

661,650

8,000

833,500

82,4
25

316,
925

3

Cash

234,500

6,000

322,925

93,8
25

In-kind

93,825

93,825

82,4
25

410,
750

Subtotal

328,325

6,000

416,750

1,27
1,70
0

Grand Total

Cash

916,330

355,
370

20,00
0

1,291,70
0

375,
300

In-kind Contribution

375,300

375,300

1,64
7,00
0

Total

1,291,6
30

355,
370

20,00
0

1,667,00
0

o Status Table of Participating Researchers' Labor Costs (First Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Choi Beom-jin

Director

Internal

Existing

173,588

6

10

8,600

8,600

Participating Researcher (Existing)

Kim Jung-woo

Manager

Internal

Existing

61,503

6

20

6,105

6,105

Lee Kyung-soo

Department Head

Internal

Existing

87,400

6

15

6,550

6,550

Choi Seung-bum

Department Head

Internal

Existing

146,500

6

11

8,050

8,050

Lee Hye-jung

Director

Internal

Existing

146,500

6

11

8,050

8,050

Lee Jae-man

Department Head

Internal

Existing

80,000

6

30

12,000

12,000

Hong Hyo-jin

Director

Internal

Existing

150,000

6

19

14,200

14,200

Kim Tae-heon

Group Leader

Internal

Existing

99,100

6

28

13,870

13,870

Park Seo-ri

Deputy Manager

Internal

Existing

65,600

6

50

16,400

16,400

Subtotal (D)

93,825

93,825

Um Sang-hyun

Director

Internal

New

70,000

6

100

35,000

35,000

Participating Researcher (New)

Seo Min-gyu

Manager

Internal

New

49,000

6

100

24,500

24,500

Scheduled Hire 1

-

Internal

50,000

5

100

20,833

20,833

New (Youth 1)

Scheduled Hire 2

-

Internal

50,000

5

100

20,833

20,833

New (Youth 2)

Scheduled Hire 3

-

Internal

New

50,000

5

100

20,833

20,833

Scheduled Hire 4

-

Internal

New

50,000

5

100

20,833

20,833

Scheduled Hire 5

-

Internal

New

50,000

5

100

20,833

20,833

Scheduled Hire 6

-

Internal

New

50,000

5

100

20,833

20,833

Scheduled Hire 7

-

Internal

New

35,000

4

100

11,666

11,666

Scheduled Hire 8

-

Internal

New

35,000

4

100

11,666

11,666

Subtotal (E)

207,830

207,830

Total (G=D+E+F)

207,830

93,825

301,655

o Status Table of Participating Researchers' Labor Costs (Second Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Choi Beom-jin

Director

Internal

Existing

173,588

12

10

17,000

17,000

Participating Researcher (Existing)

Kim Jung-woo

Manager

Internal

Existing

61,503

12

20

12,300

12,300

Lee Kyung-soo

Department Head

Internal

Existing

87,400

12

15

13,110

13,110

Choi Seung-bum

Department Head

Internal

Existing

146,500

12

11

16,100

16,100

Lee Hye-jung

Director

Internal

Existing

146,500

12

11

16,100

16,100

Lee Jae-man

Department Head

Internal

Existing

80,000

12

30

24,000

24,000

Hong Hyo-jin

Director

Internal

Existing

150,000

12

19

28,500

28,500

Kim Tae-heon

Group Leader

Internal

Existing

99,100

12

28

27,740

27,740

Park Seo-ri

Deputy Manager

Internal

Existing

65,600

12

50

32,800

32,800

Subtotal (D)

187,650

187,650

Um Sang-hyun

Director

Internal

New

70,000

12

100

70,000

70,000

Participating Researcher (New)

Seo Min-gyu

Manager

Internal

New

49,000

12

100

49,000

49,000

Scheduled Hire 1

-

Internal

50,000

12

100

50,000

50,000

New (Youth 1)

Scheduled Hire 2

-

Internal

50,000

12

100

50,000

50,000

New (Youth 2)

Scheduled Hire 3

-

Internal

New

50,000

12

100

50,000

50,000

Scheduled Hire 4

-

Internal

New

50,000

12

90

45,000

45,000

Scheduled Hire 5

-

Internal

New

50,000

12

90

45,000

45,000

Scheduled Hire 6

-

Internal

New

50,000

12

90

45,000

45,000

Scheduled Hire 7

-

Internal

35,000

12

100

35,000

35,000

New (Youth 3)

Scheduled Hire 8

-

Internal

35,000

12

100

35,000

35,000

New (Youth 4)

Subtotal (E)

474,000

474,000

Total (G=D+E+F)

474,000

187,650

661,650

o Status Table of Participating Researchers' Labor Costs (Third Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Choi Beom-jin

Director

Internal

Existing

173,588

6

10

8,600

8,600

Participating Researcher (Existing)

Kim Jung-woo

Manager

Internal

Existing

61,503

6

20

6,105

6,105

Lee Kyung-soo

Department Head

Internal

Existing

87,400

6

15

6,550

6,550

Choi Seung-bum

Department Head

Internal

Existing

146,500

6

11

8,050

8,050

Lee Hye-jung

Director

Internal

Existing

146,500

6

11

8,050

8,050

Lee Jae-man

Department Head

Internal

Existing

80,000

6

30

12,000

12,000

Hong Hyo-jin

Director

Internal

Existing

150,000

6

19

14,200

14,200

Kim Tae-heon

Group Leader

Internal

Existing

99,100

6

28

13,870

13,870

Park Seo-ri

Deputy Manager

Internal

Existing

65,600

6

50

16,400

16,400

Subtotal (D)

93,825

93,825

Um Sang-hyun

Director

Internal

New

70,000

6

100

35,000

35,000

Participating Researcher (New)

Seo Min-gyu

Manager

Internal

New

49,000

6

100

24,500

24,500

Scheduled Hire 1

-

Internal

50,000

6

100

25,000

25,000

New (Youth 1)

Scheduled Hire 2

-

Internal

50,000

6

100

25,000

25,000

New (Youth 2)

Scheduled Hire 3

-

Internal

New

50,000

6

90

22,500

22,500

Scheduled Hire 4

-

Internal

New

50,000

6

90

22,500

22,500

Scheduled Hire 5

-

Internal

New

50,000

6

90

22,500

22,500

Scheduled Hire 6

-

Internal

New

50,000

6

90

22,500

22,500

Scheduled Hire 7

-

Internal

35,000

6

100

17,500

17,500

New (Youth 3)

Scheduled Hire 8

-

Internal

35,000

6

100

17,500

17,500

New (Youth 4)

Subtotal (E)

234,500

234,500

Total (G=D+E+F)

234,500

93,825

328,325

B. Name of Joint Research and Development Institution: Stunning Box

(Unit: Thousand Won)

Research and Development Cost

Year

Support Fund Outside Research Development Expenses 5」

Research
Inducement
Calculation
Standard
Amount6」

Direct Cost

Indirect
Cost

Total

Research Facilities and Equipment Expenses

Labor Cost

Student Labor Costs

Research
Material
Cost

Commissioned
Research
Development
Cost

International Joint Research and Development Expenses

Research and Development Burden Expenses

Research Activity Expenses

Research Allowance

Subtotal

General

Special Case

General

Special Case

197,50
0

20,00
0

99,42
5

316,9
25

1

Cash

6,000

322,925

93,82
5

In-kind

93,825

93,825

291,32
5

20,00
0

99,42
5

410,7
50

Subtotal

6,000

416,750

389,00
0

25,00
0

206,8
50

620,8
50

2

Cash

25,000

645,850

187,65
0

187,6
50

In-kind

187,650

576,650

25,000

206,850

808,500

Subtotal

25,000

833,500

197,500

10,000

100,425

307,925

3

Cash

15,000

322,925

93,825

In-kind

93,825

93,825

291,325

10,000

100,425

401,750

Subtotal

15,000

416,750

784,00
0

55,00
0

406,7
00

1,245
,700

1,291,70
0

Grand Total

Cash

46,000

375,30
0

375,3
00

In-kind

375,300

1,159,
300

55,00
0

406,7
00

1,621
,000

1,667,00
0

Total

46,000

o Status Table of Participating Researchers' Labor Costs (First Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Jeong Sumin

CEO

Internal

Existing

110,000

6

98.9

54,395

54,395

Participating Researcher (Existing)

Eom Junhyeon

Director

Internal

Existing

40,000

6

98.6

19,720

19,720

Ma Jaemin

Manager

Internal

Existing

40,000

6

98.55

19,710

19,710

Subtotal (D)

93,825

93,825

Lee Junu

Department Head

Internal

New

60,000

6

100

30,000

30,000

Participating Researcher (New)

Cho Min

Department Head

Internal

New

40,000

6

100

20,000

20,000

Kang Hyojeong

Manager

Internal

New

45,000

6

100

22,500

22,500

Yun Inhee

Manager

Internal

New

45,000

6

100

22,500

22,500

Scheduled Hire 1

Director

Internal

New

84,000

5

100

35,000

35,000

Scheduled Hire 2

Department Head

Internal

New

81,000

5

100

33,750

33,750

Scheduled Hire 3

Manager

Internal

New

55,000

5

100

22,916

22,916

Scheduled Hire 4

Deputy Manager

Internal

26,000

5

100

10,833

10,833

New (Youth 5)

Subtotal (E)

197,500

291,325

Total (G=D+E+F)

197,500

93,825

291,325

o Status Table of Participating Researchers' Labor Costs (Second Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Jeong Sumin

CEO

Internal

Existing

110,000

12

98.9

108,790

108,790

Participating Researcher (Existing)

Eom Junhyeon

Director

Internal

Existing

40,000

12

98.6

39,440

39,440

Ma Jaemin

Manager

Internal

Existing

40,000

12

98.55

39,420

39,420

Subtotal (D)

187,650

187,650

Lee Junu

Department Head

Internal

New

60,000

12

53

31,800

31,800

Participating Researcher (New)

Cho Min

Department Head

Internal

New

40,000

12

53

21,200

21,200

Kang Hyojeong

Manager

Internal

New

45,000

12

100

45,000

45,000

Yun Inhee

Manager

Internal

New

45,000

12

100

45,000

45,000

Scheduled Hire 1

Director

Internal

New

84,000

12

100

84,000

84,000

Scheduled Hire 2

Department Head

Internal

New

81,000

12

100

81,000

81,000

Scheduled Hire 3

Manager

Internal

New

55,000

12

100

55,000

55,000

Scheduled Hire 4

Deputy Manager

Internal

26,000

12

100

26,000

26,000

New (Youth 5)

Subtotal (E)

389,000

576,650

Total (G=D+E+F)

389,000

187,650

576,650

o Status Table of Participating Researchers' Labor Costs (Third Year) ※ Compiled yearly for the entire project period

Personnel Classification

Name

Position (Indicate Student if applicable)

Internal/External and Support Classification*

New/Existing

Total Salary (A)

Participation Period (B, months)

Participation Rate (%) (C)

Total ((A/12)×B×(C/100))

Cash

In-kind

Total

Jeong Sumin

CEO

Internal

Existing

110,000

6

98.9

54,395

54,395

Participating Researcher (Existing)

Eom Junhyeon

Director

Internal

Existing

40,000

6

98.6

19,720

19,720

Ma Jaemin

Manager

Internal

Existing

40,000

6

98.55

19,710

19,710

Subtotal (D)

93,825

93,825

Lee Junu

Department Head

Internal

New

60,000

6

59

17,700

17,700

Participating Researcher (New)

Cho Min

Department Head

Internal

New

40,000

6

59

11,800

11,800

Kang Hyojeong

Manager

Internal

New

45,000

6

100

22,500

22,500

Yun Inhee

Manager

Internal

New

45,000

6

100

22,500

22,500

Scheduled Hire 1

Director

Internal

New

84,000

6

100

42,000

42,000

Scheduled Hire 2

Department Head

Internal

New

81,000

6

100

40,500

40,500

Scheduled Hire 3

Manager

Internal

New

55,000

6

100

27,500

27,500

Scheduled Hire 4

Deputy Manager

Internal

26,000

6

100

13,000

13,000

New (Youth 5)

Subtotal (E)

197,500

291,325

Total (G=D+E+F)

197,500

93,825

291,325

2. Detailed Usage Plan for Research and Development Expenses

(1) Research Facilities and Equipment Construction Plan - Not Applicable

(2) Purchase Costs for Research Materials (Reagents, Materials, etc.)

Research and Development Institution

Purchase Year

Purchase Quantity (g)

Purchase Cost (thousand won)

Purchase (Receipt) Date

Research Materials

Purpose of Use

Usage Destination

Establishment and Fine-tuning of Basic Trade and Export Specialized Data for AI Translation and Analysis Model Learning in the First Year, Global Customs, Export Regulation, and Buyer Specialized Dataset

Stunning Box Co., Ltd.

Year 2026

Domain-Specific AI Learning Global Trade and Export Company Dataset

1 set

20,000

August 2026

AI Model Research Laboratory / Training Server

2027

25,000

January 2027

2028

10,000

January 2028

(2-1) Research Material Production Costs (Test Products, Facilities, etc.) Usage Plan - Not Applicable

(3) Research Activity Costs (Outsourcing Fees)

Research and Development Institution

Required Costs
(thousand KRW)

Year

Service Name

Service Description

Service Company

Service Period

Deliverables

㈜0000

2026

2026

(3-1) Research Activity Costs (Intellectual Property Creation Activities, External Expert Technology Usage Fees, Software Usage Fees, Laboratory Operation Expenses, Comprehensive Management Fees, Travel Expenses, and Other Costs)

Research and Development Institution

Name
(Category)

Usage
(Company)

Quantity
(Cases)

Required Costs
(thousand KRW)

Installation Location
(Deliverables)

Year

Method

Usage (Execution) Period

PCN Co., Ltd.

2026

Entrusted Settlement Fees
(Other Costs)

Purchase

Settlement Fees for Year 2026

1

3,000

’26.7 ~12

PCN

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

Rental

Cloud Usage Fees

6

50,000

’26.7 ~12

PCN

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

5

2,000

’26.7 ~12

PCN

Test Certification Fees
(External Expert Technology Usage Fees)

Purchase

Test Analysis and Certification

2

5,000

’26.7 ~12

PCN

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Technical Consultation

2

20,000

’26.7 ~12

PCN

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

Academic Conference Participation Fees

4

15,000

’26.7 ~12

PCN

Overseas Travel Expenses
(Travel Expenses)

Purchase

Overseas Travel Expenses

1

10,000

’26.7 ~12

PCN

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

10

1,670

’26.7 ~12

PCN

2027

Entrusted Settlement Fees
(Other Costs)

Purchase

Settlement Fees for Year 2027

1

3,000

’27.1 ~12

PCN

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

Rental

Cloud Usage Fees

12

85,000

’27.1 ~12

PCN

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

5

2,000

’27.1 ~12

PCN

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Technical Consultation

2

20,000

’26.7 ~12

PCN

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Model Verification

1

10,000

’27.1 ~12

PCN

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

Conference Presentation

1

10,000

’27.1 ~12

PCN

Overseas Travel Expenses
(Travel Expenses)

Purchase

Overseas Travel Expenses

1

10,000

’26.7 ~12

PCN

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

10

6,850

’27.1 ~12

PCN

2028

Entrusted Settlement Fees
(Other Costs)

Purchase

Settlement Fees for Year 2028

1

3,000

’28.1 ~6

PCN

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

Rental

Cloud Usage Fees

6

30,000

’28.1 ~6

PCN

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

5

2,000

’28.1 ~6

PCN

Test Certification Fees
(External Expert Technology Usage Fees)

Purchase

Test Analysis and Certification

2

5,000

’28.1 ~6

PCN

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

UX Testing

2

30,000

’28.1 ~6

PCN

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Promotional Video Production Service

1

5,000

’28.1 ~6

PCN

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

Workshop Hosting Costs

5,000

’28.1 ~6

PCN

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

10

1,425

’28.1 ~6

PCN

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

Stunning Box Co., Ltd.

2026

Rental

Translation/Interpretation Model Training/Inference
Utilization GPU Cloud

6

60,000

’26.7 ~12

StunningBox

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

-

2,000

’26.7 ~12

StunningBox

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Domain-Specific Review Expert

-

23,000

’26.7 ~12

StunningBox

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

AI Technology Seminar

5

3,925

’26.7 ~12

StunningBox

Overseas Travel Expenses
(Travel Expenses)

Purchase

Overseas Travel Expenses

10

3,000

’26.7 ~12

StunningBox

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

10

2,000

’26.7 ~12

StunningBox

Meeting Costs
(Meeting Costs)

Purchase

Meeting

10

1,500

’26.7 ~12

StunningBox

Report Costs
(Other Costs)

Purchase

Report Binding, Printing, etc.

10

3,000

’26.7 ~12

StunningBox

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

2027

Rental

Translation/Interpretation Model Training/Inference
Utilization GPU Cloud

12

130,000

’27.1 ~12

StunningBox

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

5

2,000

’27.1 ~12

StunningBox

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Domain-Specific Review Expert

-

33,000

’27.1 ~12

StunningBox

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

Researcher Training and Training Expenses

5

3,925

’27.1 ~12

StunningBox

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

10

1,000

’27.1 ~12

StunningBox

Meeting Costs
(Meeting Costs)

Purchase

Meeting

10

1,500

’27.1 ~12

StunningBox

Report Binding, Printing, etc.
(Other Costs)

Purchase

Report Binding, Printing, etc.

10

3,000

’26.7 ~12

StunningBox

Cloud Usage Fees
(Cloud Computing Service
Usage Fees)

2028

Rental

Translation/Interpretation Model Training/Inference
Utilization GPU Cloud

6

62,000

’28.1 ~6

StunningBox

Office Supplies Expenses
(Laboratory Operation Costs)

Purchase

Office Supplies Purchase

5

1,000

’28.1 ~6

StunningBox

Test Certification Fees
(External Expert Technology Usage Fees)

Purchase

Test Analysis and Certification

2

15,000

’28.1 ~6

StunningBox

Expert Usage Fees
(External Expert Technology Usage Fees)

Purchase

Domain-Specific Review Expert

-

22,000

’28.1 ~6

StunningBox

Conference and Seminar Participation Fees
(Research Personnel Support Fees)

Purchase

Researcher Training and Training Expenses

3

5,425

’28.1 ~6

StunningBox

Domestic Travel Expenses
(Travel Expenses)

Purchase

Domestic Travel Expenses

5

1,000

’28.1 ~6

StunningBox

Meeting Costs
(Meeting Costs)

Purchase

Meeting

5

1,000

’28.1 ~6

StunningBox

Report Binding, Printing, etc.
(Other Costs)

Purchase

Report Binding, Printing, etc.

10

3,000

’28.1 ~6

StunningBox

(4) Indirect Costs

Research and Development Institution

Required Costs
(thousand KRW)

Year

Details

Detailed Information

2026

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

6,000

PCN Co., Ltd.

2027

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

8,000

2028

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

6,000

2026

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

6,000

Stunning Box Co., Ltd.

2027

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

25,000

2028

Achievement Utilization Support Costs

Intellectual Property Rights Application and Registration

15,000

3. Research Fund Procurement and Maintenance Plan

7. Utilization Plan and Expected Effects of R&D Achievements (Commercialization Goals and Strategies, etc.)

1) Commercialization/Product (Service) Goals and Plans

The commercialization goal of this project is to create a market centered on 'AI Translation SaaS', where SMEs can automate global sales, translation, and documentation without separate IT investment by linking their data. By occupying the empty position between existing general-purpose translation AI (DeepL, Google) and high-cost professional translation, it blocks entry into the competitive exclusive domain based on the Stanning Box 1.1 million+ corpus (including 500,000 regulatory certification).

(1) Stage-wise Commercialization Roadmap – Four-Stage Market Entry Strategy

Sales
Target

Stage

Timing

Commercialization Goal

Key Execution Strategy

KPI

Pilot Completion of 3 Customer Companies.
PMF Verification.
Acquisition of TTA/KAIC Official Certification

In R&D
(~'28.6)

Free Demonstration→ Securing Reference.
Weaponization of Sales with Quantitative Introduction Effect Data

-

Stage 1 Base
Construction

Adopted by 3 Companies, Daily Work Reduced by 2+ Hours, Certification Acquired

Simultaneous Deployment of Triple Channels (Government Support + Association + App Store).
30%+ Free Tier→Paid Conversion Rate

'28.7~
'29.6

Official Launch of SaaS. AI Voucher, KOTRA Channel Deployment. 50+ B2B Paid Clients

6 billion

Phase 2 Commercialization

50+ Paid Clients, MRR
500 million, NPS 50+

Opening of Industry-specific Marketplace.
Cafe24 · Naver App Store. 15 Languages.
500+ Companies

Bulk Contracts with Industry Groups.
Expansion of API B2B2B. Upselling to Existing
Customers with Regulatory Certification Package

'29.7~
'31.6

13 billion

10%+ B2B Market Share,
LTV/CAC>3

Stage 3 Expansion

Entry into Japan, Southeast Asia, and Middle East.
Local CRO partners.
eCTD Global. 1,000+ companies

API export model. Overseas localization (Japanese, Arabic, Vietnamese enhancement). On-premises enterprise

Stage 4 Global

'31.7~
'33.6

22 billion

Overseas sales 45%+,
1,000+ companies

(2) Product (Service) Line-up and Pricing Strategy

Form

Price Range

Differentiation Points / Strategic Role

Product Name

Target Customers

Natural transition from Free Tier (100 uses/month). Remove entry barriers. High-volume inflow channels

Export beginner companies,
Individual traders, E-commerce sellers

SaaS subscription (self-service)

100,000 KRW per month (1.2 million KRW/year)

StunningBiz AI Starter

Includes custom onboarding for company RAG. Unlimited on
StunningBiz AI Business
line interpretation. Main revenue source (50% of total)
Dedicated CSM allocation. Enhanced security (on-

StunningBiz
premise). Custom model. High AI Enterprise
Lock-in·LTV

SaaS subscription (account manager)

300,000 KRW per month (3.6 million KRW/year)

Export growing companies
(annual exports 100 million to 1 billion KRW)

500,000+ KRW per month
(6 million+ KRW/year)

Mid-sized companies with 1 billion+ KRW exports, global companies' Korean branches

SaaS + on-premises hybrid

CE
300,000 to 1 million
eCTD
1 to 5 million

Per-use billing (standalone or SaaS add-on)

Companies pursuing regulatory approval
(health supplements, cosmetics, electronics)

Exclusive access to StunningBox with 500,000 corpora.
50-70% less compared to existing RA agencies.
High margin 80%+

Regulatory certification package (premium)

Cafe24·Naver·Amazon·CRM·ERP operators

API call usage-based pricing

500,000 KRW base + 20 to 100 KRW per call

Viral expansion as an app within the platform. Revenue proportional to user growth. Network effect

API B2B2B

(3) Market Entry Strategy – Triple Channel Simultaneous Deployment
(Securing 150 companies within 1 year of commercialization)

Goal to secure within 1 year

Execution Plan

Channel

Target

① Government support AI voucher 20 companies + export voucher 10 companies = 30 companies

Registered as AI voucher provider (up to 40 million KRW/company for free). Export voucher·Global promotion

Export beginner companies. Initial non-charge with government support

Link with export support projects. KOTRA overseas trade mission local experience

Zero cost

MOU with Trade Association·KCCI·Industry-specific Export Cooperation. 20% group discount. Free seminars·Demo days once a month. Utilize references from 3 demand companies

Export growing companies. Ensure trust through association recommendation

② Industry organization
B2B

15 companies through associations + 5 directly = 20 companies

Registration in Cafe24·Naver Smart Store·Amazon Seller Central App Store. Viral spread with 'automated translation' app.
1-Click installation. Large inflow through Free Tier

E-commerce sellers (direct overseas sales). Minimizing entry barriers as an app

③ Platform
B2B2B

Including Free Tier, 100 companies
(Paid conversion target of 30%+)

※ Goal to secure 150 companies including Free Tier and 50+ paid conversions through three channels within 1 year of commercialization.

Achieve BEP by second year (at 300 company subscription point). The regulatory certification package is a non-competitive area based on StunningBox's exclusive data, serving as a high-margin (80%+) premium source of income.

2) Commercialization Strategy (Clarity of Business Model (BM), Marketing and Sales Channel Strategy, etc.)

(1) Business Model (BM) Structure — Quadruple Billing System and Strategic Roles

Expected Margin

Sales Contribution (%)

Revenue Model

Billing Method

Price Range

Customer Transition Path

Strategic Role

Stable recurring revenue
(ARR). Growth engine. Churn rate <
5%/month managed

Free Tier →
Starter →
Business →
Enterprise

Monthly fixed + exceeded usage charges

100,000 to 500,000 KRW/month

55~65%

50%

BM1. SaaS
Subscription

Exclusive high-margin cash cow.
Entry barrier through StunningBox
with 500,000 corpora

Per-use billing (document type×country)

CE 300,000 to 1 million
eCTD 1 to 5 million

80%+

25%

SaaS customer
Upselling + direct sales

BM2. Regulatory
Certification Package

High conversion barriers
(Lock-in). Average contract 24 months+.
High LTV

Implementation
1 to 5 million
maintenance monthly
100,000 to 300,000

Implementation cost + monthly
maintenance fee

60~70%

15%

Enterprise
customers target
consulting sales

BM3. RAG Custom
Onboarding

Platform network
Effect. Growth proportional to usage. Viral
expansion

API call usage-based
pricing

500,000 KRW base + 20 to 100 KRW per call

70~75%

10%

App store installation
→ automatic billing

BM4. API
B2B2B

(2) Core Indicators of Profitability (Unit Economics)

Year 1 Target

Year 3 Target

Basis of Calculation

Indicator

Year 1: High CAC (Customer Acquisition Cost) due to free inflow from government channels. Year 3: Increase in organic inflow through viral + app store. Regulatory certification upselling + enterprise ARPU (Average Revenue Per User) naturally increases as proportion rises. Churn rate improves from 5% to 3% monthly, reflecting ARPU increase as LTV (Customer Lifetime Value) contract duration extends. SaaS industry profitability benchmark of LTV / CAC ratio is over 3x—expected to be greatly exceeded from the outset. Monthly fixed cost of approximately 150 million won basis. MRR BEP (Break-even Point) 150 million = 300 subscriptions (weighted average ARPU of 500 thousand)

500,000 won/company

300,000 won/company

300,000 won

450,000 won

7.2 million won (24 months)

16.2 million won (36 months)

14.4 times

54 times

End of 2nd year (300 subscriptions)

-

(3) Marketing Strategy — Stage-wise Investment Allocation and Key Tactics

Investment Proportion

Years 1-2 (Market Penetration Phase)

Year 3 and beyond (Efficiency Phase)

Marketing Domain

30%

Blogs and YouTube targeting export practitioners: practical content such as 'Consult with Buyers Without Interpreters', 'Cut Regulatory Certification Costs by 50%'. Three case studies of demand companies (Case

SEO Assetization → Over 50% Organic Inflow. Accumulation of 30+ Industry Success Stories. Regular webinars twice a month

Content Marketing

videos/articles production

Focus on keywords for Google Ads and Naver SA such as 'Export Translation', 'Interpretation AI'. Retargeting (encourage conversion from Free Tier users to paid). LinkedIn B2B lead generation

25%

Concentration on channels achieving over 300% ROAS. Maintain CAC below 300,000 won. AI-based bid automation

Performance Advertising

MOU with 20 KOTRA overseas trade centers → local export companies experience. Regular seminars at Trade Association. Nationwide demo tour at 62 export support centers. Joint promotions with Cafe24 and Naver

30%

Secure 10 reseller partners by industry. Partner revenue commission model. Partnerships with overseas CRO and RA agencies

Partner/Channel Sales

KOTRA's news release on export success stories. AI conference (ACL, IWSLT) paper presentations. Open-source model release → Viral marketing in the HuggingFace community

15%

Establish 'K-AI Export Solution' brand positioning. PR in international media (e.g., TechCrunch Asia)

PR/Branding

(4) Market Strategy — Online, Offline, Global Triad

Channel

Detailed Execution Plan

Target Number of Customers

Revenue Contribution Proportion

Direct registration on stunningbiz.ai website. Free Tier trial → Self-service upgrade to paid. Automated card payment and invoice issuance. 24-hour chatbot CS

Year 3: 250 companies

Online Direct Sales 40%

Cafe24 App Store, Naver Smart Store App, Amazon Seller Central plugins. Minimize entry barriers with 1-Click installation. Automatic payment and settlement within platform

Year 3: 150 companies

App Store Integration 25%

Dedicated sales team (3 members) for Enterprise. Group contracts linked with industry associations/export cooperatives. Sales to similar companies based on demand company references. Regulatory certification package consulting sales

Year 3: 80 companies

B2B Direct Sales 25%

Japan: IT distributor partnerships. Southeast Asia: KOTRA overseas trade center collaborative local experience programs. Middle East: Partnership with RA agencies specializing in Halal/Islamic regulations. API exports (OEM)

Year 4: 200+ companies

Global Channel 10% → 45%

※ Core Market Strategy: Achieve high-volume low-cost inflow via online direct sales + app store during Years 1-2 → Expand high-value Enterprise and regulatory certification sales through B2B direct sales in Year 3 → Achieve 45%+ overseas revenue contribution in Year 4 with global channels. Optimize overall marketing cost from 35% of sales (Years 1-2) to 20% (Year 3 and beyond) as content assets and partner channels become the organic growth engine.

3) Disclosure Level and Diffusion Strategy of Development Results

To balance the public interest and commercial profitability of government R&D outcomes, this project adopts an 'Open Core' strategy. AI models and datasets are made publicly available to enable free use by SMEs and research communities, while core business logic (Regulatory Certification Engine, HITL Verification, Billing) is provided only as SaaS to secure commercialization profits.

(1) Disclosure Level by Development Result — Open Core Strategy

Development Result

Disclosure Level

Disclosure Method and Utilization Plan

Reason for Non-disclosure / Commercialization linkage

Published on HuggingFace Hub under Apache 2.0 license. Includes industry-specific variant models (Manufacturing/Retail/Food/IT). SMEs can deploy directly on their servers for free utilization

Full Disclosure

Lack of RAG, RLHF, Monitoring in Self-operation vs. SaaS → Maintain SaaS Value-add

Business Translation AI Model Weights

Contribute to Academic Community + Strengthen K-AI Brand. SaaS differentiated with Speaker Separation, RAG, Plugins

Full Disclosure

Two-way Streaming Interpretation Model for Korean ↔ 8 Languages. Inference Code + Docker Image Release. First Domestic E2E Korean Simultaneous Interpretation Open Source

E2E Speech Interpretation Model Weights

100,000 out of 1.1 million disclosed (9%). The rest retained as non-disclosed to maintain SaaS model quality differentiation

Maximum Disclosure

Business Parallel Corpus of more than 100,000 sentence pairs published on AI Hub. Full disclosure of industry-specific trade terminology dictionaries (3 types). De-identified data completed

Business Parallel Corpus (Training Data)

[Non-disclosure] Hybrid Reranker Weights, Auto-boarding Logic by Company RAG → SaaS Key Differentiation Factors

Selective Disclosure

[Disclosure] RAG Vector Search Engine Core, Example API Integration Code (Python, Node.js), Docker Distribution Image, Developer Documentation. GitHub Open Source Repository Maintenance

RAG Search Engine · API Integration Code

Non-disclosure

Provided only as a premium SaaS service. Exclusive engine based on 500,000 Stuttering Box Corpus. Accessible only via API (charged per action)

Regulatory Certification Formatting Engine (eCTD, CE, FCC)

Core High-margin (80%+) Commercialization Area. Block Competition Entry. Protect 10 years of Accumulated Data Assets

Provided only within SaaS platform. Quality Monitoring, A/B Testing, Automatic Re-learning Pipeline, Usage Billing System

Non-disclosure

HITL Verification and SaaS Core Infrastructure. Billing and Dashboard Non-disclosure underpins Profit Model

(2) Prior Establishment of Rights Relationships for Data and AI Results
Origin data, fine-tuning models, and common platform rights are divided into

three, and the method of transferring model weight ownership to demand companies is not adopted in accordance with public R&D result management principles.

(2-1) Ownership Plan by Deliverable

Deliverables

Content

Attribution Entity

Source data provided by the demand enterprise (document, voice, consultation
source data
powers, etc.)

Attributed to the demand enterprise

Joint Outcomes

1) The enterprise-specific weight for the demand enterprise is classified as joint outcomes for
fine-tuning model (enterprise-specific), the demand enterprise holds free usage rights within the scope of the business purpose
Common platform, base model
Integrated platform common infrastructure and base model are system·API Attribution to the leading·joint research institution

Attributed to the Leading Joint Institution

(2-2) Assured Provisions to Reflect in Contract Annexes

Content

Attribution Entity

Data provided by the demand enterprise may only be used within the scope of this business's purpose,
Provision Purpose/Usage Scope
Prior written consent is required for reuse or secondary learning
Retention period, specify retention period and disposal procedure after project completion in the Data Provision Assurance Agreement

Usage Rights Granted The fine-tuning results can be used freely by the demand enterprise within the scope of this business purpose
Deletion·Relearning Procedure Define in advance data deletion procedures and relearning reflection procedures upon the demand enterprise's request

(3) Outcome Diffusion Strategy — 4 Main Channels

Detailed Execution Plan

Goals (within 3 years)

Expected Effect

Diffusion Channel

Establishment of K-AI brand. Contributions to voluntary improvements by subsequent researchers and enterprises. Increase in SaaS reputation

HuggingFace Hub model released + GitHub code release. Model card, benchmark, usage guide provided. Community operation for issue management, PR review. Quarterly releases

GitHub Star 1,000+, downloads 5,000+/month, 20 contributors+

Open Source Community

Revitalization of domestic NLP research. Solving the problem of lacking translation data for SMEs. Returning government R&D outcomes

Register 100,000+ sentence pairs of business parallel corpora on AI Hub. Release 3 terminology dictionaries by industry. Obtain NIA data quality certification. Available for use as subsequent R&D training data

AI Hub downloads 3,000+ cases, utilization papers 10+

Public Data Contribution

Verification of technical reliability. Connection to subsequent R&D. Evidence of TRL improvement. Global technology leadership

Publication of papers at top AI conferences like ACL, EMNLP, and IWSLT (2+ papers). Publication at domestic conferences (KCC, Korean Information Science Society). 3 patent applications+(RAG, E2E interpretation, regulated formats). Technical reports available

2+ papers, 3
patents+, and 2
technical reports

Academic·Technical Presentations

Public Disclosure of Reports

Core channel for diffusion to 500+ SMEs
Key channel for distribution. Connection to subsequent government support projects.
Expansion of export SME base

Distribution to 20 KOTRA overseas trade centers (free experience).
Registration as AI voucher suppliers (up to 40 million won/company free).
Distribution of demos to 62
trading associations and export support centers. Register excellent case studies with the Ministry of SMEs and Startups

Government·Institutional
Related Diffusion

AI Voucher 50+ companies,
200 KOTRA experiences
62 export support centers distribution

(4) Free Tier Continuous Operation — Guarantee Access for SMEs

Contents of Free Tier Offerings

Item

Expansion Benefit Conditions

100 translations per month + 60 minutes of voice interpretation + 5 automatic document generations. 1GB of enterprise RAG DB. 1,000 API calls/ month. No time limit (permanently free)

Basic Free Offering

30-day free trial added upon
paid transition

Expanded Benefits for SMEs

Enterprises with a confirmation of SME: Double the free limit (200 uses/month+120 minutes). Up to 40 million won of free usage when linked to AI vouchers. 50% discount for first-year export beginner companies

Submission of SME certificate or export performance certificate

Unlimited free API for universities, research institutes, non-profit organizations. Separate educational dashboard provided. Joint improvement agreement for the model in joint research

Submission of certificate for educational/research purpose. Prohibition of commercial resale

Free Provision for Research·Education

※ Core strategy for public disclosure: 'Open the model, close the service' — Ensure accessibility for SMEs and community trust through open-source disclosure of AI models and datasets, while maintaining profitability by keeping core SaaS differentiation elements such as regulation certification engines, RAG automation, HITL quality management proprietary. This strategy follows the verified BM model of global open-core companies like Red Hat, Elastic, and MongoDB.

4) Expected Effects (growth potential of target market, entry feasibility, projected sales, export substitution, etc.)

(1) Growth Potential of Target Market

'24 Scale

'27 Outlook

'30 Outlook

CAGR

Market Segmentation

Source

Grand View Research 2024

MarketsandMarkets
2024

Global AI Translation Market

1.8 billion dollars

3.5 billion dollars

7 billion dollars

30%+

Global Real-time Interpretation Market

500 million dollars

1.2 billion dollars

2.8 billion dollars

33%+

Domestic AI Translation Market

120 billion won

280 billion won

650 billion won

32%+

KISTI·IITP estimation

KOTRA Export Enterprise DB

Number of Domestic Export SMEs (Potential Clients)

About 100,000 companies

-

-

-

→ The global AI translation and interpretation market is experiencing ultra-rapid growth with CAGR of 30%+, while the domestic market is expected to

reach 650 billion won by 2030. A 1% market share (1,000 companies) among the potential 100,000 clients can achieve annual revenue of 22 billion won

(2) Market Entry Feasibility — Basis for Competitive Advantage

Basis for Entry

Specific Details

Level of Advantage Compared to Competitors

DeepL·Google: No business-specific corpora. Domestic competitors: No regulated corpora. → 1~2 years of entry barriers

Monopoly Data Asset

110 million+ parallel corpora in Stunning Box (50 million+ regulated 60 million business). Impossible for competitors to replicate the exclusive data and professional translation asset accumulated over 15 years in a short time

E2E Technology Preemption

Based on InfiniSST, E2E real-time interpretation (within 2 seconds) has no domestic commercialization examples. Existing

No domestic competitors (Blue Ocean).
Google S2ST does not support Korean.

Fundamental quality superiority over Cascade (STT→MT→TTS)

→ Technological lead for at least 1 year

Startup competitors: Lack of SaaS operational performance, unverified stability. → Overwhelming enterprise credibility

PCN with 25 years in IT services (Revenue 27 billion KRW, 170 employees). 110 billion KRW court SI order. Operating the Oz-i AI platform in a commercial setting. Holds GS-certified API platform.

Verified SaaS infrastructure

Minimization of initial customer acquisition cost (CAC). Possible to bring in 50 companies+ annually through government channels for free → Basis for break-even point (BEP) in 2nd year

AI voucher (up to 40 million KRW/company free) + KOTRA 20 overseas trade centers + Trade Association + 62 Export Support Centers. Secured references from 3 client companies

Securing government channels

(3) Projected Revenue and Job Creation

Classification

'28 (1st Year)

'29 (2nd Year)

'30 (3rd Year)

'31 (4th Year)

'32 (5th Year)

Domestic sales 12 billion KRW
Export sales (API·Overseas
SaaS) 10 billion KRW
Total sales 22 billion KRW
Operating profit 7.7 billion KRW (35%)
Accumulated subscribed companies 1,000+
Accumulated employment (direct) 51 people
5-year accumulated sales 68 billion KRW

4 billion KRW

6 billion

8 billion KRW

10 billion KRW

2 billion KRW

3 billion KRW

5 billion KRW

8 billion KRW

6 billion

9 billion KRW

13 billion

18 billion KRW

900 million KRW (15%)

2 billion KRW (22%)

3.6 billion KRW (28%)

5.8 billion KRW (32%)

150 companies

300 companies

500 companies

800 companies

16 people

25 people

34 people

42 people

※ Basis for sales estimation: SaaS ARPU 300,000 KRW/month (weighted average), annual price increase of 20%, improvement of churn rate from 5% to 3% monthly, regulatory certification average unit price 1.5 million KRW·1,000 cases annually+

(4) Export Replacement Amount (Technology Export + Import Replacement Effect)

'28

'29

'30

'31

'32

Export Replacement Items

Direct export of overseas SaaS·API 10 billion KRW
Replacement of overseas translation outsourcing cost
6 billion KRW
Total export replacement amount 16 billion KRW
5-year accumulated export replacement amount 44 billion KRW

2 billion KRW

3 billion KRW

5 billion KRW

8 billion KRW

1 billion KRW

2 billion KRW

3 billion KRW

4 billion KRW

3 billion KRW

5 billion KRW

8 billion KRW

12 billion KRW

※ Overseas SaaS·API export: Entry into Japan·Southeast Asia·Middle East + Export of OEM API. Import replacement: Replacement effect of existing overseas translation services (DeepL Pro·Specialized translation outsourcing). Basis for calculation of export replacement: Average annual translation outsourcing cost 35 million KRW per exporting company × number of replacement companies

※ Summary: This project, in a CAGR 30%+ high-growth market, possesses commercialization potential to achieve cumulative sales of 68 billion KRW, export replacement of 44 billion KRW, and direct employment of 51 people within 5 years, based on exclusive data (1.1 million corpora) + E2E technology lead + verified SaaS infrastructure, with a very high investment efficiency (sales/support fund = 34x) compared to government support of 2 billion KRW.

8. Implementation Plan for Safety and Security Measures in R&D

1) Safety Measures Implementation Plan

During the R&D execution process, adhere to safety rules regarding the use of research facilities, equipment, and computing equipment, and provide regular safety education to participating researchers. Conduct

pre-checks before equipment installation and operation, and periodically identify potential risk factors such as electricity, fire, and equipment overheating. In the event of an accident, establish an emergency contact network and response procedures to enable immediate reporting and response, and systematically manage safety inspection results and training history.

2) Security Measures Implementation Plan

Sensitive information such as research data, AI training data, demand company's products, prices, contracts, and export documents is categorized and managed according to importance. Encryption is applied to data storage and transmission sections, and access is controlled to ensure that only authorized personnel can access the data through role-based access management and user authentication. Access, viewing, and modification logs are recorded and regularly inspected, and external transfer is prevented by checking the person in charge's approval and de-identification status.

3) Implementation Plan for Other Measures

Data containing personal information and corporate secrets is clarified in terms of collection purpose and usage scope, and utilized in research after de-identification and anonymization processing. Participating researchers comply with pledges on research ethics, personal information protection, and security, and establish clear rights and utilization scopes of AI training data and research outcomes in advance. Also, regularly conduct self-inspections to verify compliance with safety, security, and ethics, and continuously manage inspection results and related supporting documents.

Attachment 1. Confirmation Form for National R&D Project Application and Execution Details

Plan for Utilization and Management of Research Lab Operating Expenses for Profit Organizations
Self-Inspection Results and Obligatory Confirmation Form for Security-Level for Application Project

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Attachment 1. Confirmation Form for National R&D Project Application and Execution Details

Confirmation Form for National R&D Project Application and Execution Details

Project Name

SME Technology Innovation Development (R&D)
Project Number

RS-2026-25606465
Project Title

Integrated Platform for Automation of Global Sales, Counseling, Translation, and Documentation for SME Export Revenue Innovation
Lead R&D Institution

PCN Co., Ltd.

In applying for the SME Technology Development Support Project, this institution submits the list of national R&D projects that are applying or executing or have executed as provided in [Attachment 1-1], confirming that the applied project is not eligible for pre-support/pre-development or duplicate support with ongoing or completed national R&D projects.

Furthermore, for verification of pre-support/pre-development status, it is confirmed that if requested by the managing agency or expert, consent to inquire about pre-support/pre-development status for the projects listed in [Attachment 1-1] is given.

If any false, forged, or omitted facts in listed details are confirmed, it is agreed that no objection will be made to actions such as cancellation of selection, termination of agreement, and participation restrictions in accordance with relevant laws.

[Attachment 1-1] List of Implementation/Application Details of National R&D Projects

July 1, 2026
Lead R&D Institution: PCN Co., Ltd.
Representative: Kwang-Hyun Song

[Attachment 1-1] List of Implementation/Application Details of National R&D Projects (All to be filled based on the date of closure of this announcement)

◦ Lead R&D Institution

Project Number

Research and Development Project Name

Lead Research and Development Institution Name

Research and Development Period (Participation Period)

Contents of Implementation (Subject, Objective, Method)

Central Administrative Agency (Specialized Agency)

Remarks (In Progress/Completed/Applying)

Name and Role of Research and Development Institution (Lead/Cooperative)

Real-time Visualization of Space Situational Awareness / Establishment of Orbit Information Comparison, Integration, and Metaverse Linkage System / Real-time Data Collection/Comparison and AI-based Orbit Simulation Linkage

RS-2023-00232134

Development of Visualization and Linkage Technology for Space Object Orbit Information Based on Metaverse

PCN Co., Ltd.

2023.04.01 ~ 2026.12.31
(2023.04.01 ~ 2026.12.31)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Lead)

Development of Emotional AI Model Generation and Training Dataset Construction Technology

RS-2026-25514284

PCN Co., Ltd.

2026.04.01 ~ 2028.12.31
(2026.04.01 ~ 2028.12.31.)

Infrastructure for Collecting Biometric Signals and Development of Intelligent Annotation SaaS Platform

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Lead)

Digital Service Carbon Emission Management / AI-based Life Cycle Carbon Footprint Calculation and Management System Establishment / Data Collection from Heterogeneous Systems and Data Warehouse Analysis Infrastructure Construction

RS-2026-\n25527887

Development of Key Technology for LCI-based Digital Service Carbon Footprint Calculation and Tracking Management

EasyG Co., Ltd.

2026.04.01 ~ 2029.12.31
(2026.04.01 ~ 2029.12.31)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Joint)

Development of Character and Webtoon Copyright Opt-Out Response Technology and Evaluation Model in Generative AI Image Service Environment

Development of Generative AI Copyright Protection / Copyright Opt-Out Application and Infringement Identification Response Technology / Development of Copyright Infringement Evaluation Algorithm based on Large-scale Webtoon Data

RS-2026-\n25504479

Seokyeong University Industry-Academia Cooperation Foundation

2026.04.01 ~ 2028.12.31
(2026.04.01 ~ 2028.12.31)

Korea Creative Content Agency

In Progress

PCN Co., Ltd. (Participant)

Development of AI Agent-based XR Telepresence Platform for Context Awareness of Remote Multi-party Conversations

XR-based Remote Collaboration Platform / Construction of Multilateral Conversation Context Awareness and XR Interaction Platform / Development of Multimodal Data Synchronization and Persona-reflecting Multi-avatar Technology

RS-2026-\n25518171

PCN Co., Ltd.

2026.04.01 ~ 2029.12.31.
(2026.04.01 ~ 2029.12.31)

Korea Creative Content Agency

In Progress

PCN Co., Ltd. (Lead)

Comprehensive Investigation System for Drug Crimes / Dark Web / Establishment of Integrated Analysis Environment for Virtual Asset Crime Data / Development of Unstructured Data Collection, Knowledge Graph Loading, and Investigator Visualization Dashboard

RS-2026-\n25535833

Development of Integrated Drug Investigation System Linking Dark Web and Virtual Asset Transaction Tracking

Sungkyunkwan University Industry-Academia Cooperation Foundation

2026.04.01 ~ 2028.12.31
(2026.04.01 ~ 2028.12.31)

Science Security Promotion Center

In Progress

PCN Co., Ltd. (Participant)

Commercialization of Drone Video-based XR Management / Ultra-low Latency Drone Video Visualization and AI Situation Awareness Integrated Management System / P5G-based Transmission and VLM/LLM-based On-site Briefing Technology Application

RS-2026-\n25561688

Drone Video Management XR Goggle System for Disaster Response

PCN Co., Ltd.

2026.05.01 ~ 2027.12.31
(2026.05.01 ~ 2027.12.31)

Korea Institute for Advancement of Technology

In Progress

PCN Co., Ltd. (Lead)

PCB Manufacturing Process Optimization / Advancement of AI Visual Twin and Mass Production Line Monitoring Platform Construction / Core Process Layout Design and Equipment Data Management Pipeline Construction

RS-2026-\n25567830

Development of Semiconductor PCB Industry-specific Manufacturing Problem-solving Multi AI Agent Technology

InterX Co., Ltd.

2026.04.01 ~ 2028.12.31
(2026.04.01 ~ 2028.12.31)

Korea Institute for Advancement of Technology

In Progress

PCN Co., Ltd. (Participant)

RS-2024-\n00339583

Development and Demonstration of XOps-based AI Policy Simulation Platform for Addressing Social Issues in Population Decline Areas

PCN Co., Ltd.

2024.04.01 ~ 2027.12.31
(2024.04.01 ~ 2027.12.31)

Simulation for Solving Community Issues / Construction of DataOps/MLOps-based AI Policy Simulation Platform / Development of Multi-institutional Data Linking and Geospatial Information-based Change Monitoring UI

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Lead)

Rendering Server-based XR Content / Development of High-Resolution XR Content Offloading Technology in Cloud-Native Environment / Development of HW-accelerated Rendering and Pre-warming/Load Balancing Technology

RS-2024-\n00354327

Development of High-Quality Ultra-Low Latency XR Cloud Computing Technology

PCN Co., Ltd.

2024.04.01 ~ 2026.12.31
(2024.04.01 ~ 2026.12.31)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Lead)

Enhanced Data Security Federated Computing / Establishment of FedAI Central Server and Neural Network to Solve Data Silos / Design of Federated Learning Orchestration and Data Non-sensitization Module Development

RS-2024-\n00460980

Development and Demonstration of Distributed Data Federated Computing Technology for Enhanced Data Utilization

Korea Institute of Science and Technology

2024.07.01 ~ 2028.12.31
(2024.07.01 ~ 2028.12.31)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Participant)

Development of Workplace-contextual Understanding and Inference Capability Giant AI Model-based Robot Manipulation Task Learning Technology

RS-2024-\n00442513

Sejong University Industry-Academia Cooperation Foundation

2024.07.01 ~ 2028.03.31
(2024.07.01 ~ 2028.03.31)

AI Model for Robot Manipulation Learning / Development of VLM-based Robot Task Cognition and Synthetic Data Generation Platform / Automatic Generation and Learning Simulation of Random Synthetic Data based on Digital Twin

Korea Institute for Advancement of Technology

In Progress

PCN Co., Ltd. (Participant)

AI-based Equipment Autonomous Control and State Measurement Technology Development for Semiconductor PCB Business Process Optimization

RS-2024-\n00507739

BlueNex Co., Ltd.

2024.09.01 ~ 2028.12.31
(2024.07.01 ~ 2028.03.31)

Autonomous Control of PCB Equipment / Development of AAS-based Process Recipe Autonomous Control and Quality Prediction Platform / Design of Digital Twin for Demand Companies and Connection of Physical Operation Simulation

Korea Institute for Advancement of Technology

In Progress

PCN Co., Ltd. (Participant)

Development of Multi-purpose Autonomous Orchard-Agricultural Work Robot Partnering Platform for Innovation in Future Agricultural Productivity

Comprehensive Management of Autonomous Agricultural Work Robots / Demonstration of HAARS Platform for Autonomous Orchard and Agricultural Work Robots / Construction of Data-driven AI Decision Support and Robot Partnering Marketplace

RS-2025-\n25449059

PCN Co., Ltd.

2025.09.01 ~ 2028.12.31
(2025.09.01 ~ 2028.12.31)

Korea Institute for Advancement of Technology

In Progress

PCN Co., Ltd. (Lead)

Emotion Evaluation in XR Content / Development of Multimodal-based Emotion Measurement and Evaluation Indicators in XR Content / Development of Biomonitoring Patch and AI-based Emotion Recognition Evaluation Model

RS-2026-\n25516382

Development of Quantitative Evaluation Technology for Emotional and Sensory Reactions to Virtual Convergence Content

Korea Institute of Science and Technology

2026.04.01 ~ 2028.12.31.
(2026.04.01 ~ 2028.12.31.)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Participant)

Reproduction of Space-Audio for Immersive Experience / Development of 3D Virtual Space and Custom Audio Rendering based on Multimodal Data / Data Collection and Place Attribute Inference using VQA-VLM based on Geospatial Information

RS-2026-\n25522497

Development of Multi-party Immersive Experience Visualization Space-Audio Composition and Reproduction Rendering Technology

Korea Electronics Technology Institute

2026.04.01 ~ 2028.12.31
(2026.04.01 ~ 2028.12.31)

Institute for Information & Communications Technology Planning and Evaluation

In Progress

PCN Co., Ltd. (Participant)

Appendix 2. Plan for Utilization and Management of Research Laboratory Operation Costs of Profit Organizations Not Applicable

▪

military

group

special

1. Status of R&D Project

Project Title

Research and Development Project Name

Research and Development Institution Name

Research Director

Affiliation

Name

Research and Development Period

Total

20 . . . ~ 20 . . . (year month)

First Year

20 . . . ~ 20 . . . (year month)

nth Year

20 . . . ~ 20 . . . (year month)

2. Research Laboratory Operation Costs for Profit Organizations (Amount)\n(Unit: Thousand KRW)

Costs for Office Equipment and Software (A)

Costs for Equipment and Supplies to Maintain a Comfortable and Clean Environment in the Laboratory (B)

Classification

Consumable Costs Necessary for Laboratory Operation

Total\n(C=A+B)

Overall Period

(Not Available)

Stage 1

(Not Available)

nth Stage

(Not Available)

* Enter the amount planned at the time of the initial research and development project agreement signing\n** Enter costs for purchasing, installing, leasing, and using office equipment and software in A\n*** Enter costs for purchasing and maintaining equipment and supplies necessary to maintain a comfortable and clean laboratory environment in B\n**** Add or delete rows according to the annual structure of the research and development project

3. Research Laboratory Operation Costs for Profit Organizations (Items)\n(Unit: ea, Thousand KRW)

Classification

Classification

Quantity (G)

Unit Price (H)

Amount (I=G×H)

First Year

Item Name

Item Name

nth Year

Item Name

* Enter the amount planned at the time of the initial R&D project agreement
** Enter the item name of the equipment, software, or supplies in 'Item Name'
*** Enter 'Office Use' in the 'Classification' field when using (planning) costs for office equipment and software, or 'Environmental Maintenance' when using (planning) costs for equipment and supplies for maintaining heating, cooling, and cleanliness in the laboratory
**** Rows can be added or deleted according to the stage configuration of the R&D project
***** The content entered in Article 3 must be consistent with the content entered in Article 2
****** For-profit institutions must submit this document to the head of the central administrative agency when applying for an R&D project (If not submitted, retrospective application is not possible, and if the laboratory operation cost is not intended to be used, it is excluded)

Year Month Day

Research Manager: ○ ○ ○ (Seal)

Main/Joint/Subcontracted Research and Development Institution: ○ ○ ○ (Seal)

Director of Small and Medium Business Technology Information Promotion Agency

Attachment 3. Self-inspection results of the security grade of the applying project and obligation confirmation upon designation as a security project

Not Applicable

Self-inspection results of the security grade of the applying project and upon designation as a security project

Obligation Confirmation

Project Name

Project Number

Project Name

Main Research and Development Institution

Main Research Manager

Joint Research and Development Institution

Joint Research Manager

Subcontracted Research and Development Institution

Subcontracted Research Manager

Our institution confirms that upon applying for the above-mentioned task in the Small and Medium Business Technology Development Support Project, the security grade matches the attached document

Our institution has confirmed all the attached security measures for the Small and Medium Business Technology Development Support Project hereinafter referred to as 'Security Measures', and pledges that if the project applied for is confirmed as a security project, we will perform the duties

and responsibilities as a research and development institution conducting security tasks stipulated in the Security Measures

In addition, if the security grade of the above-mentioned project is designated as a 'security project' by the classification results of the security task classification committee of the professional institution during the execution of the above project, or due to changes in government policy, while it was applied as a 'general task' based on the self-inspection results, we will fulfill the duties and responsibilities as a research and development institution conducting security tasks stipulated in the Security Measures

Our institution pledges to bear sanctions and civil and criminal liability according to the National Research and Development Innovation Act and related security law provisions if we violate obligations according to the Security Measures

Attached Scheduled Task Security Grade Classification Self-Inspection Table

Small and Medium Business Technology Development Support Project Security Measures

Year Month Day

20

Main Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)
Joint Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)
Subcontracted Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)

Minister of Small and Medium Enterprises and Startups (Director of Small and Medium Business Technology Information Promotion Agency)

Attachment 1> Not Applicable

Scheduled Task Security Grade Classification Self-Inspection Table

□ Based on Article 45 of the Enforcement Decree of the National Research and Development Innovation Act (classification of security projects related to research and development tasks), please inspect. (Security grades are 'security project' and 'general project'.)
□ Tasks classified as security projects must be managed according to <Attachment 2> 'Small and Medium Business Technology Development Support Project Security Measures' and the internal security management regulations of the research and development institution conducting the security project.

Project Title

Project Number

Inspection Date

20 . . .

Project Name

Transcribe the project name entered in the project reception system as is

Main Research and Development Institution

Main Research Manager

Joint Research and Development Institution

Joint Research Manager

Subcontracted Research and Development Institution

Subcontracted Research Manager

Num
ber

Security Grade Classification and Assessment Criteria

Assessment Result

Yes

No

1

Research and development project related to the defense improvement project pursuant to Article 3(1) of the Defense Business Act

2

Research and development project related to technology under nationalization following refusal of technology transfer abroad

3

Research and development project related to future core technology recognized by the head of the central administrative agency as needing protection

4

Research and development project related to national core technology pursuant to Article 2(2) of the Act on Prevention and Protection of Industrial Technology Leakage

5

Research and development project related to technology requiring export permission or restriction under Article 19(1) of the Foreign Trade Act

6

Tasks recognized as needing to be classified as security projects for other reasons

※ If the research institution applies as a 'security project', detailed reasons must be provided

Security Grade Final Confirmation Result: □ Security Project □ General Project
※ If there is any 'Yes' in the above review results, it is classified as a security project

Our institution confirms that we have sincerely reviewed and completed all the above items.

20 Year Month Day

Main Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)
Joint Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)
Subcontracted Research and Development Institution Name:
Representative: (Seal) Manager: (Seal)